National Blockchain Framework BaaS – Rapid Prototyping Release

USER MANUAL | PART – A: ROLE MGMT AND NETWORK SETUP

NBF Team
C-DAC | WWW.CDAC.IN

Revision History

Version	Date	Author	Remarks
0.1	18-10-2023	NBF Team	Initial document

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1. National Blockchain Framework

National Blockchain Framework is an initiative of Ministry of Electronics and Information Technology (MeitY) to make India ready for large scale adoption of Blockchain and enable trust for applications in the domain of e-Governance. The framework is being offered as a service with technology stack supporting the following capabilities:

- End-to-end, rapid and secure application development
- OpenAPIs for seamless integration
- Distributed infrastructure

The framework is also capable of hosting design patterns based on Smart Contract Templates addressing requirements of various application domains. This facilitates faster development and on-boarding of new and existing applications. It enables the Government departments to develop, operate and govern a Blockchain network and e-Governance application with the required performance and security.

BasS Features:

- Faster End-to-End Blockchain Stack Setup
- Single and Multi Node Setup
- Low Code Blockchain Development
- Application Domain specific and generic Smart Contracts
- Supported Platforms: Fabric and Sawtooth
- Security across Network, Smart Contract and Application layers
- Easy onboarding of new and existing applications
- OpenAPIs for seamless integration
- Bring Your Own Infrastructure

This document helps the users to understand the complete functionality and features of BaaS.

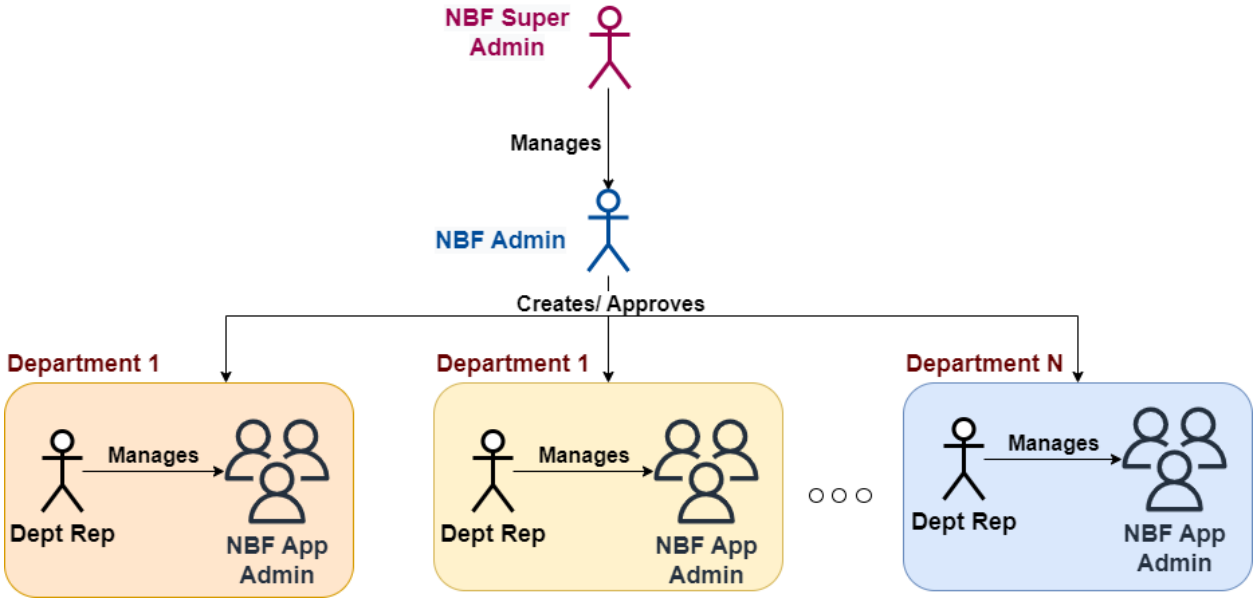
2. Roles and Responsibilities

2.1. NBF BaaS supports four Roles

1. NBF Super Admin
2. NBF Admin
3. DeptReg (Department Registration)
4. NBF App Admin

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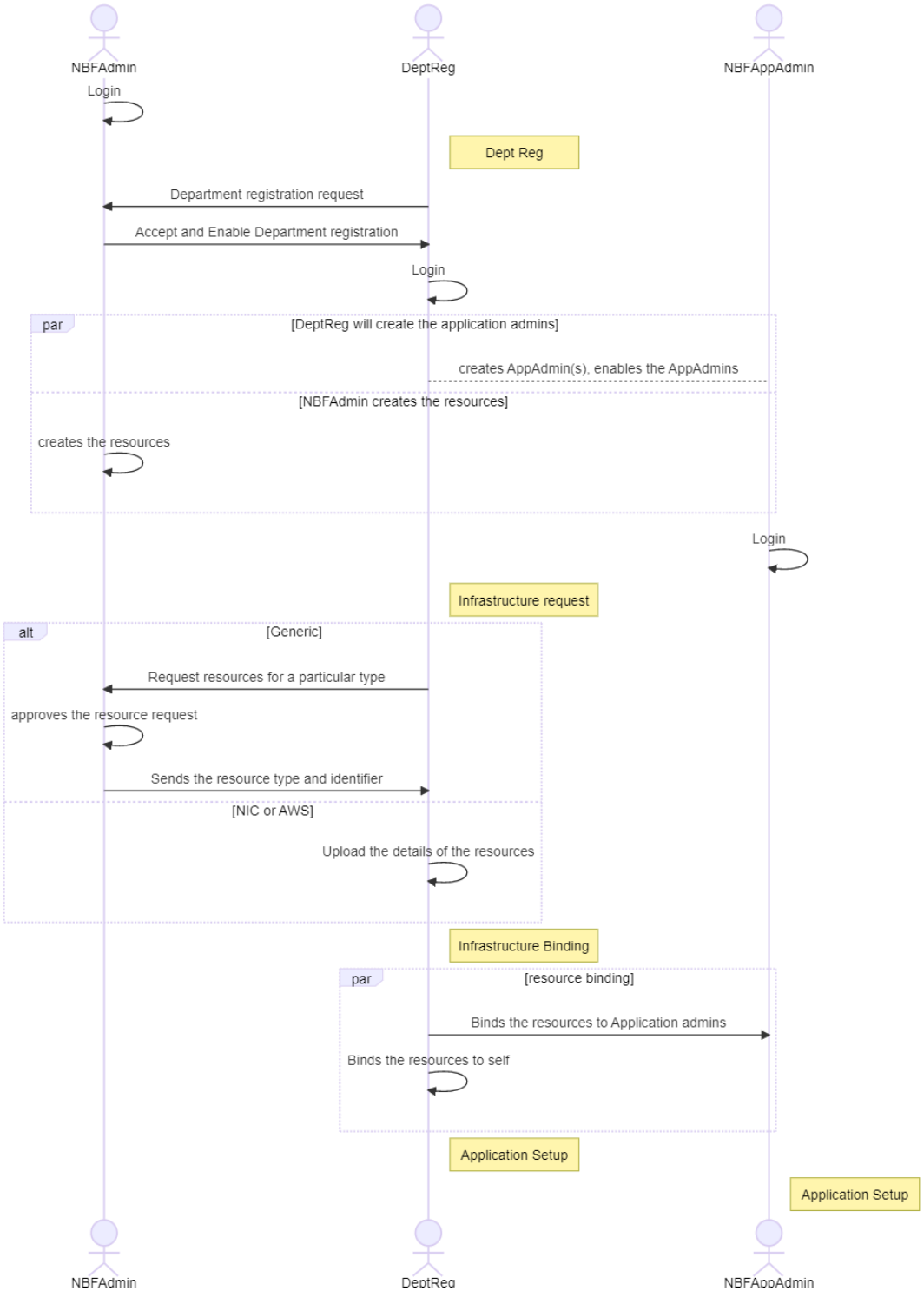
2.2. Responsibilities of each Role



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{Roles, Actions}	Super Admin	NBF Admin	DeptReg (Department Registration)	NBF App Admin
Register and Login	• Register & Login • User Management	Login (will be registered by NBF Super admin)	Register & Login – through NBF and e-Pramaan	Login (will be registered by Department user)
Requests and Approvals	•	• Approve Department registrations • Approves Infrastructure requests from multiple departments (in case of Generic Connector)	• Creates and enables NBF App Admins. • Requests for required infrastructure (Generic Connector) • Binds NBF App Admins to Infrastructure	NA
Creates	•	• Creates list of available Blockchain platforms, versions • Creates a pool of resources (in case of Generic Connector) • Creates domains, stakeholders and …	NA	NA
Blockchain Platform	NA	NA	• Network setup • Deploy smart contract • Enrol users	• Network setup • Deploy smart contract • Enrol users

2.3. Sequence diagram

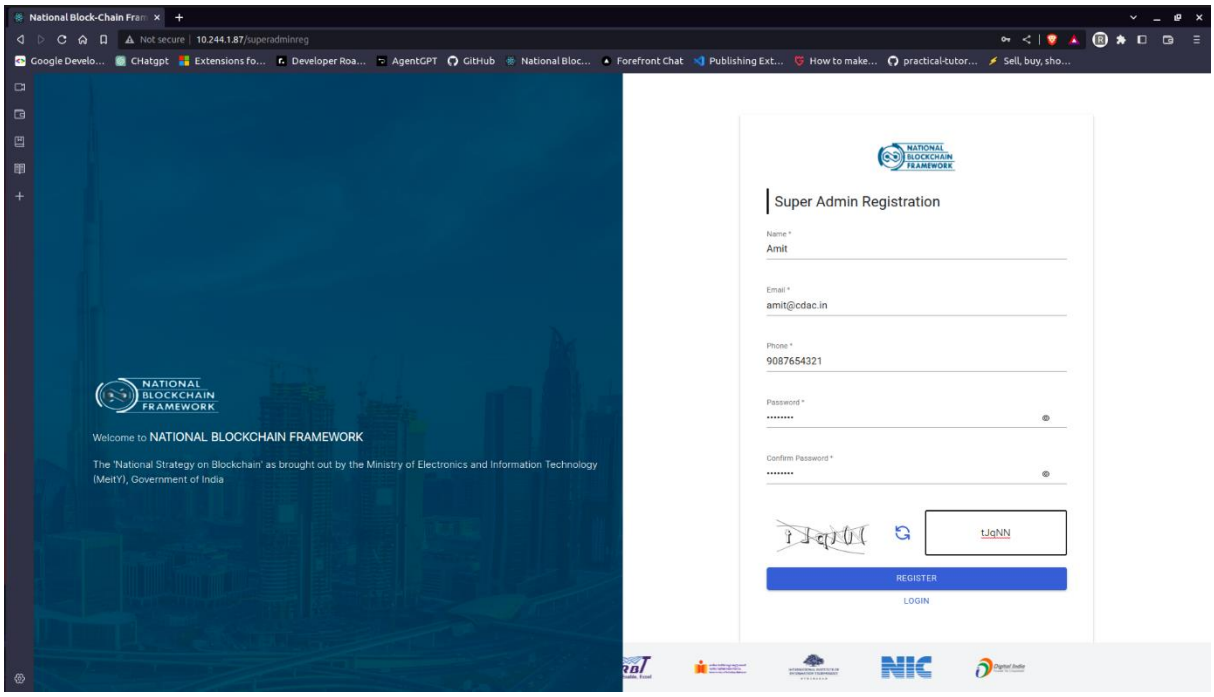


3. Role: NBF Super Admin

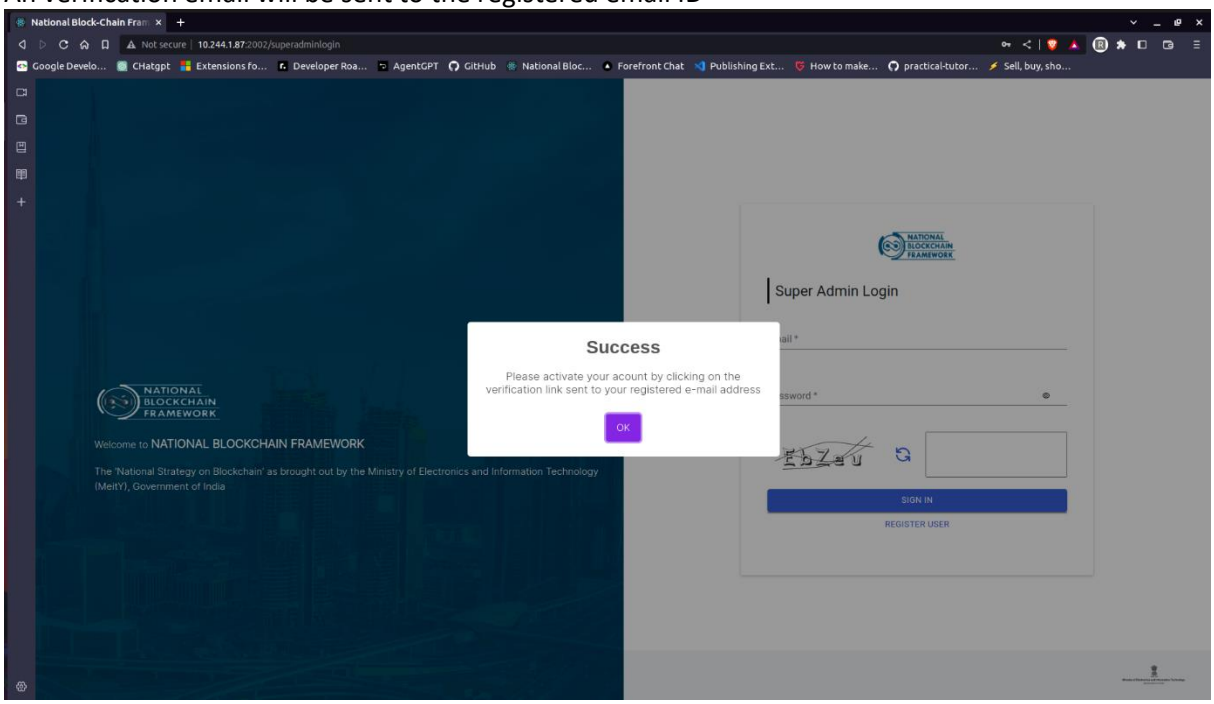
URL:<ip>/superadminreg

3.1. Register and Login

Enter the required details and click on **Register** button

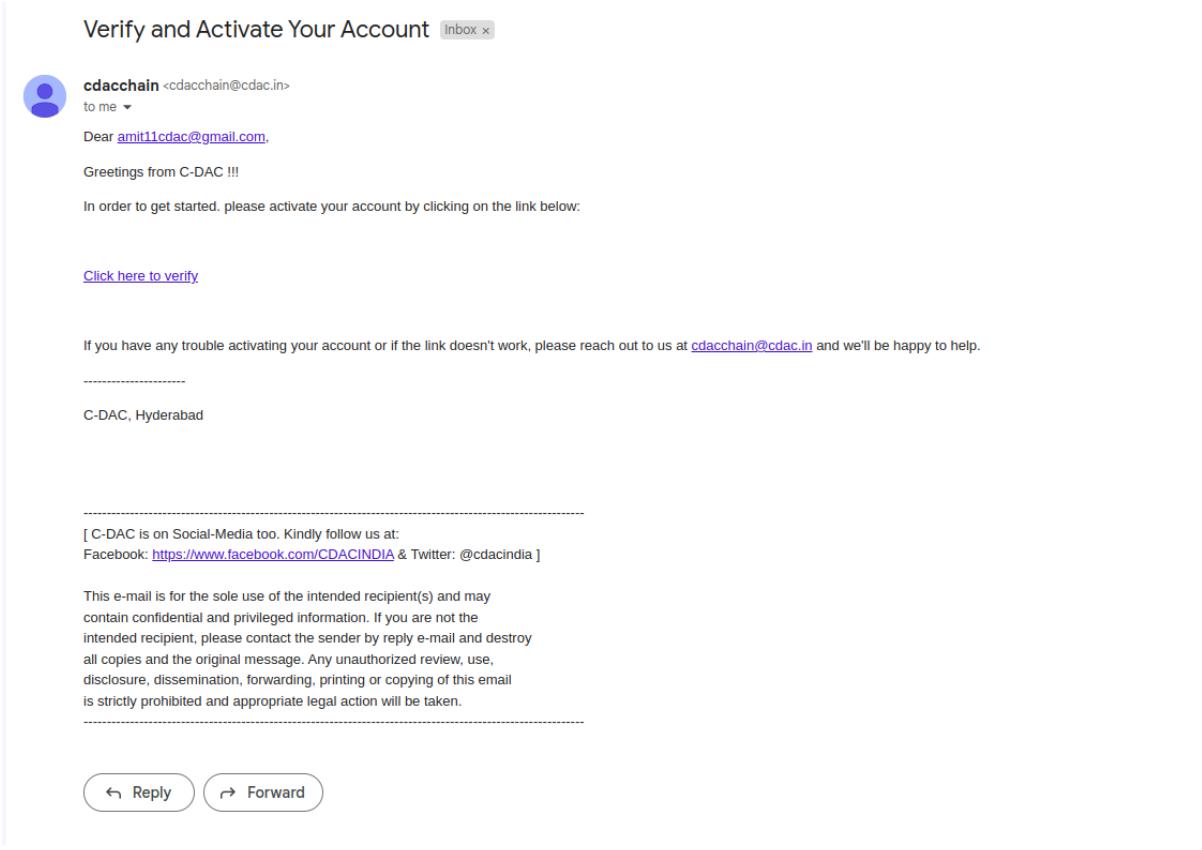


An verification email will be sent to the registered email ID



Click the link to verify

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Verify and Activate Your Account Inbox ×

cdacchain <cdacchain@cdac.in>
to me

Dear amit11cdac@gmail.com,

Greetings from C-DAC !!!

In order to get started. please activate your account by clicking on the link below:

Click here to verify

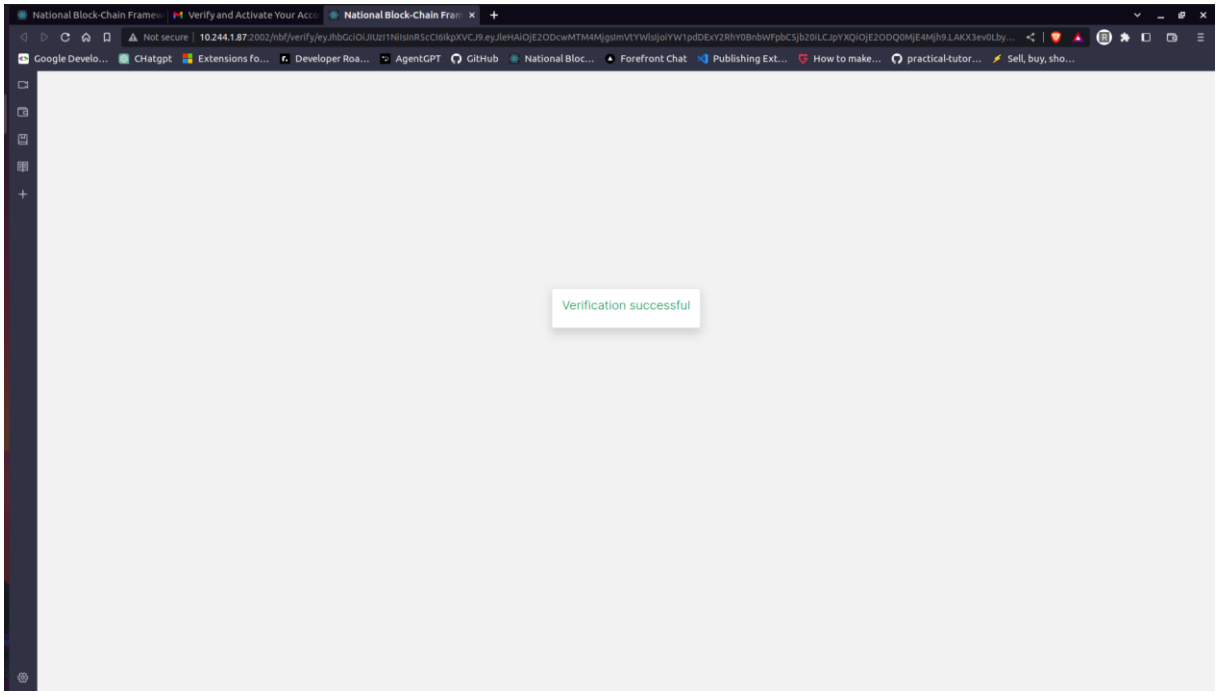
If you have any trouble activating your account or if the link doesn't work, please reach out to us at cdacchain@cdac.in and we'll be happy to help.

C-DAC, Hyderabad

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Reply Forward



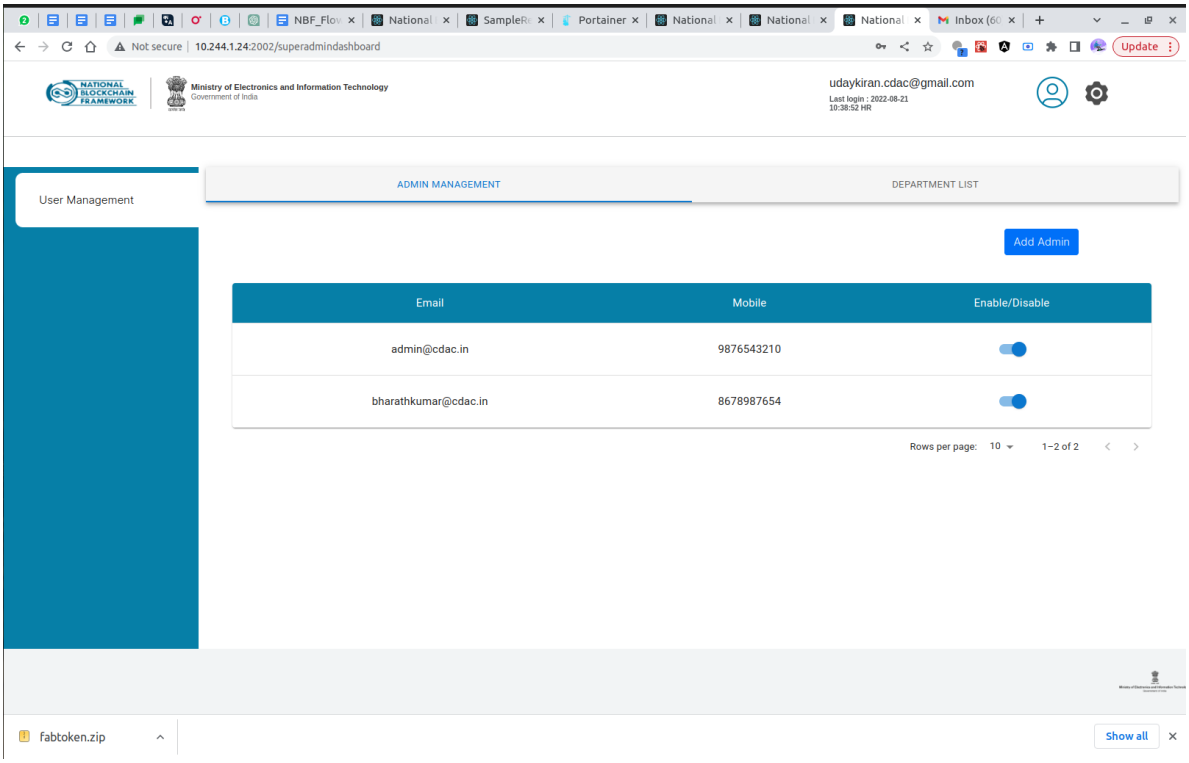
After successful Verification, superadmin can login.

NB: Only one super admin is allowed for the complete BaaS

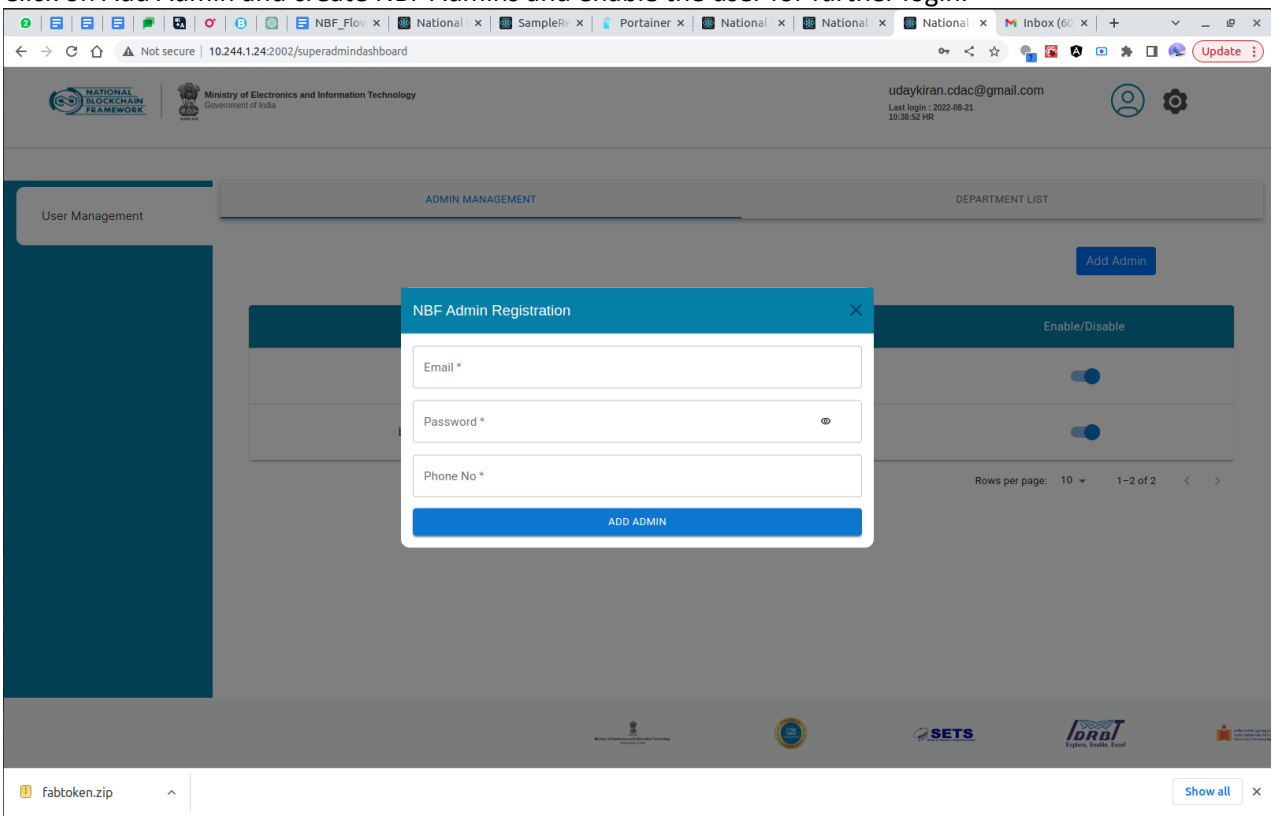
3.2. User Management

After successful login, Super admin can create Multiple NBF Admins.

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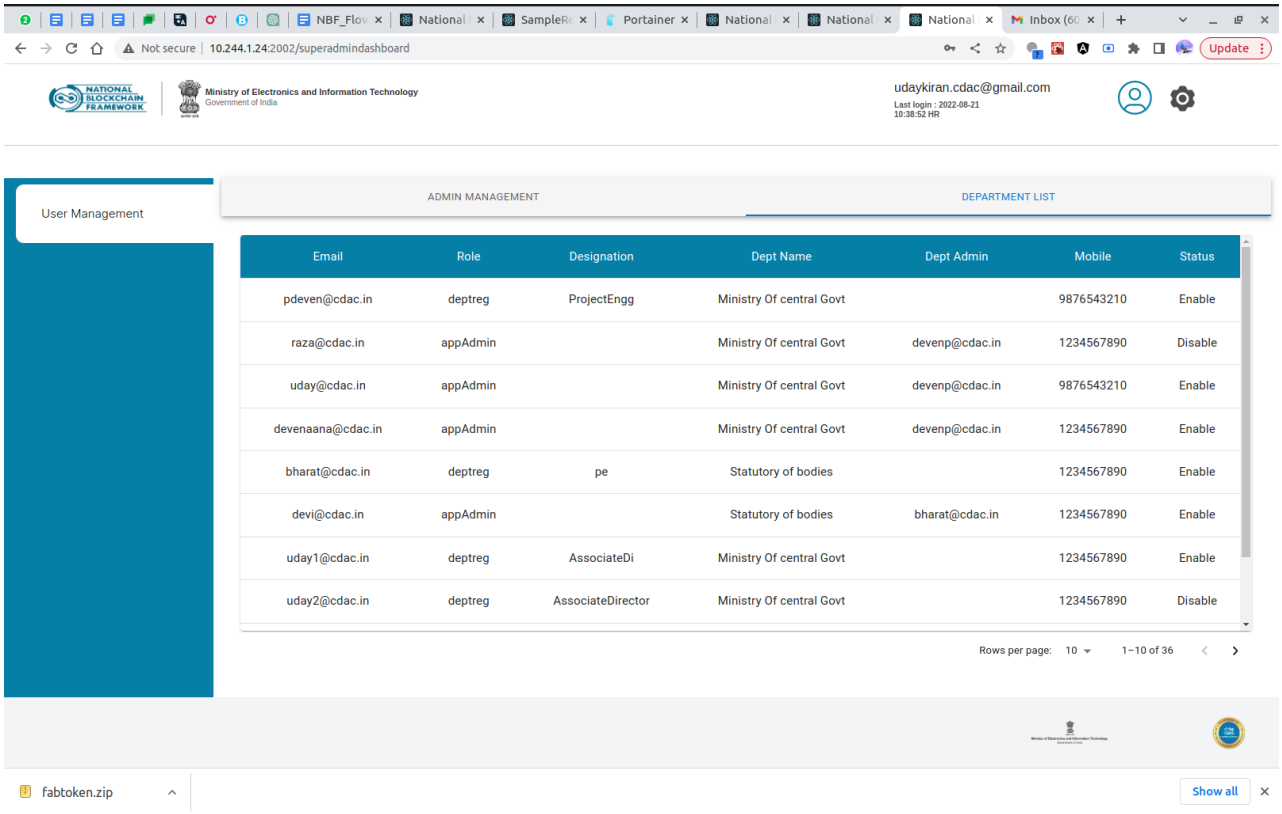


Click on Add Admin and create NBF Admins and enable the user for further login.



Super admin can view all the other users under Department List

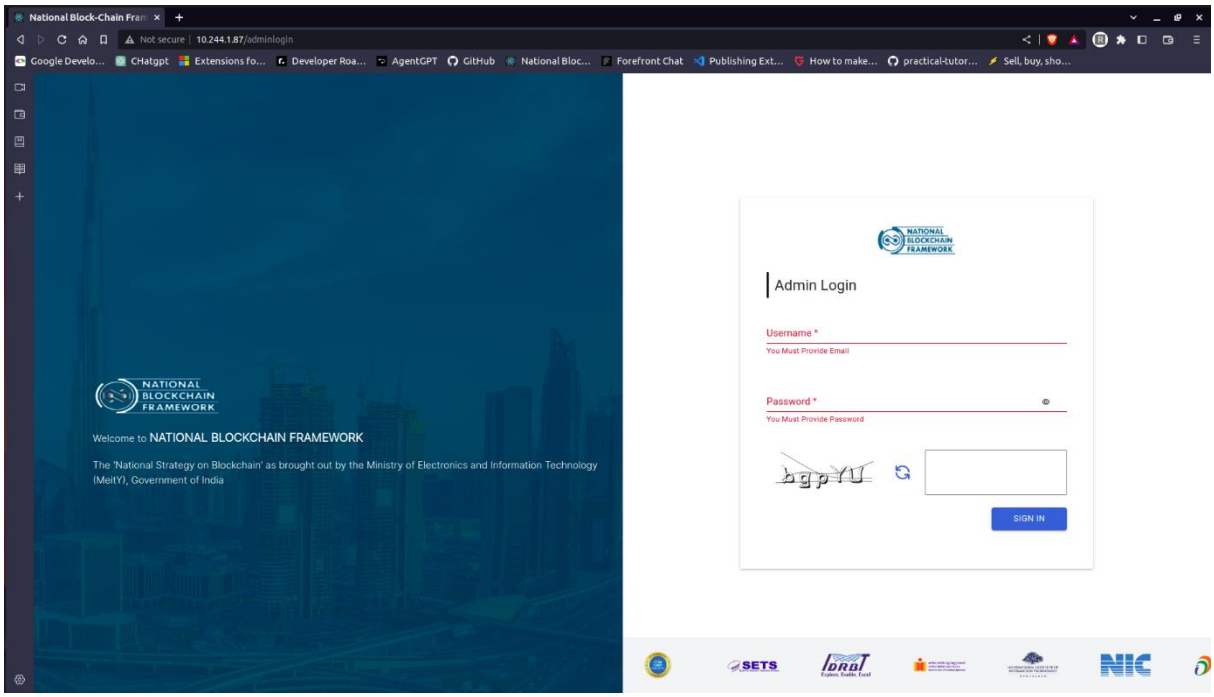
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4. Role: NBF Admin

With the login credentials created by NBF Admin, Admins can login

URL: <ip>/adminLogin



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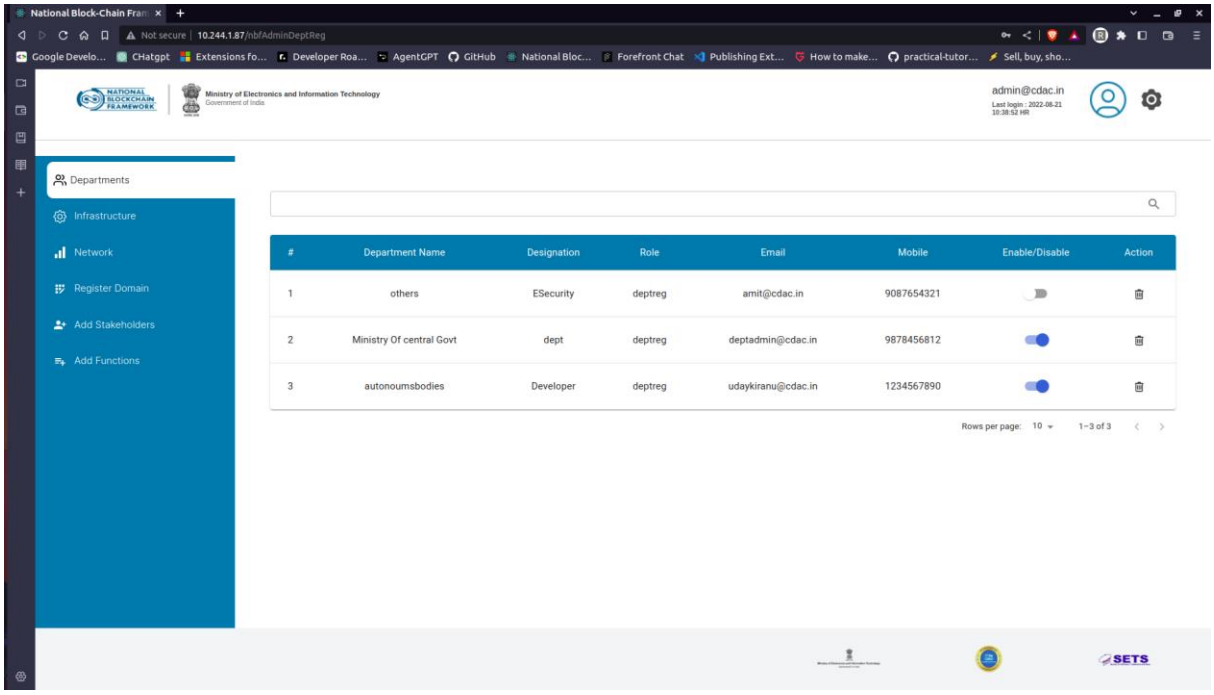
After successful Login, (s)he can do the following:

1. Approve the department registrations
2. Add Infrastructure
3. Approve the Infrastructure requests
4. Make an entry of the Blockchain Platform
5. Available Connectors (supports Bring Your Own Infrastructure)
6. Add Domains
7. Add Stakeholders
8. Add smart contracts that are available as templates for download

4.1. Departments

4.1.1. Approve Department Registrations

Click on Departments in the left panel and enable the department registrations after verifying the necessary documents.

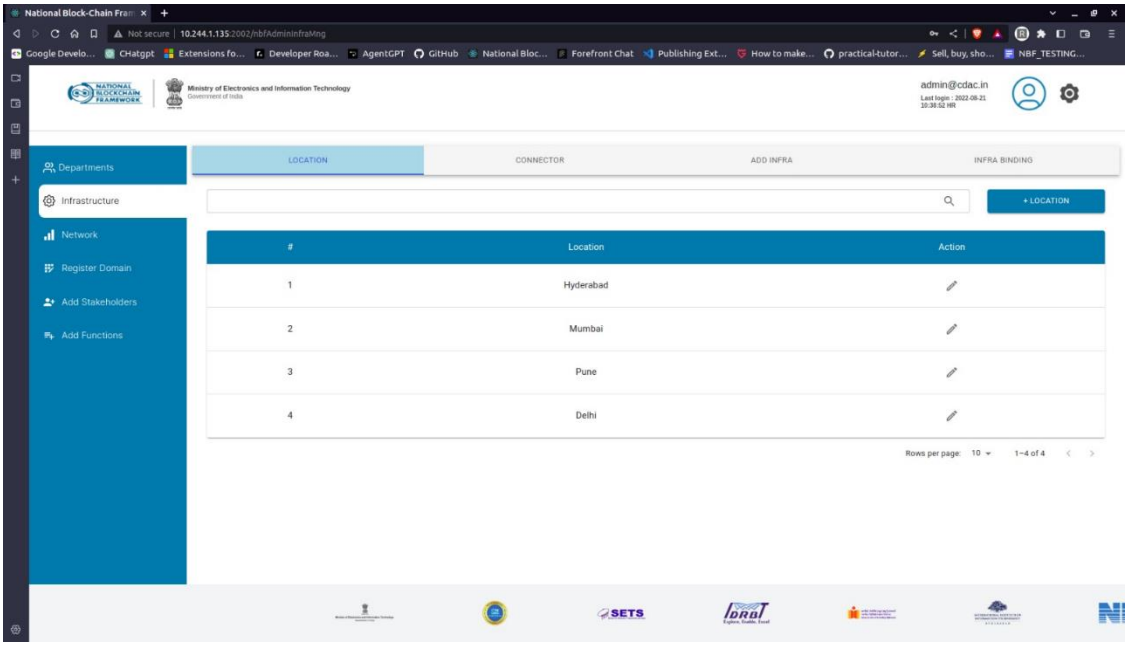


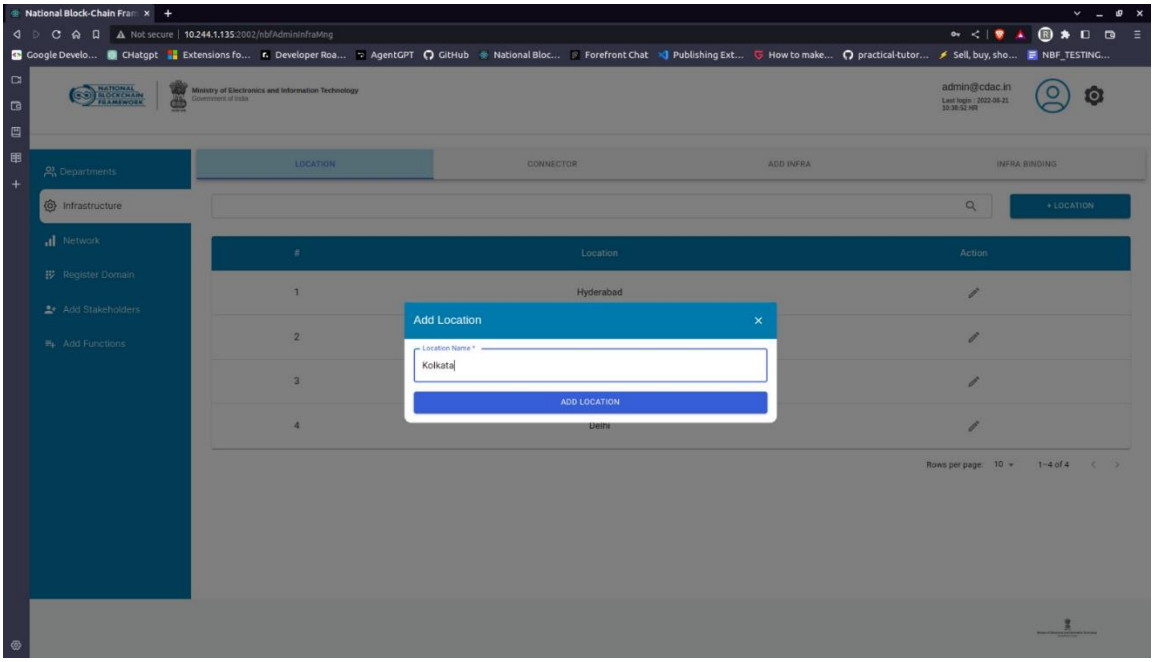
4.2. Infrastructure

4.2.1. Locations

Under the Infrastructure menu, locations where the resources are available can be captured through locations tab.

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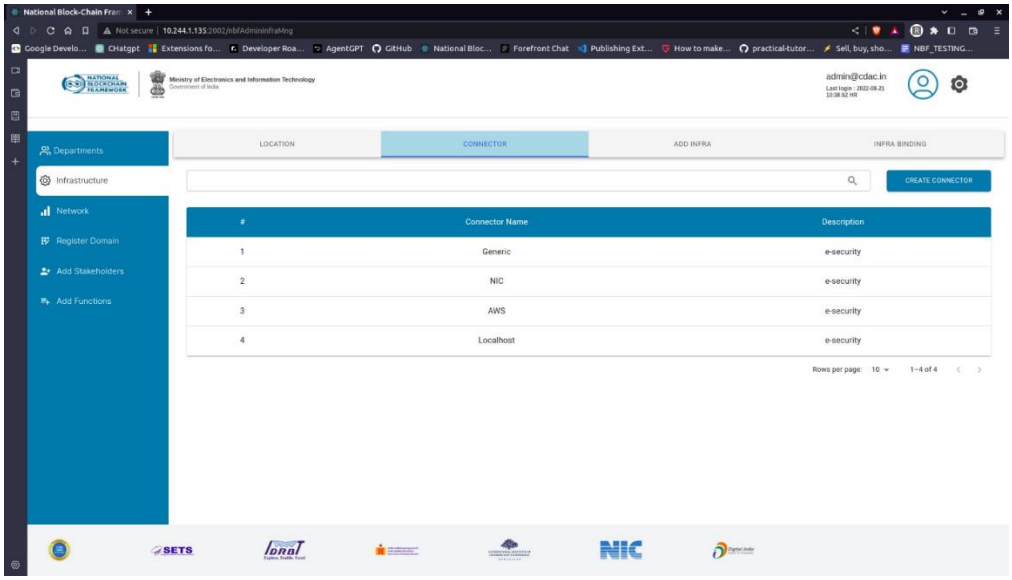


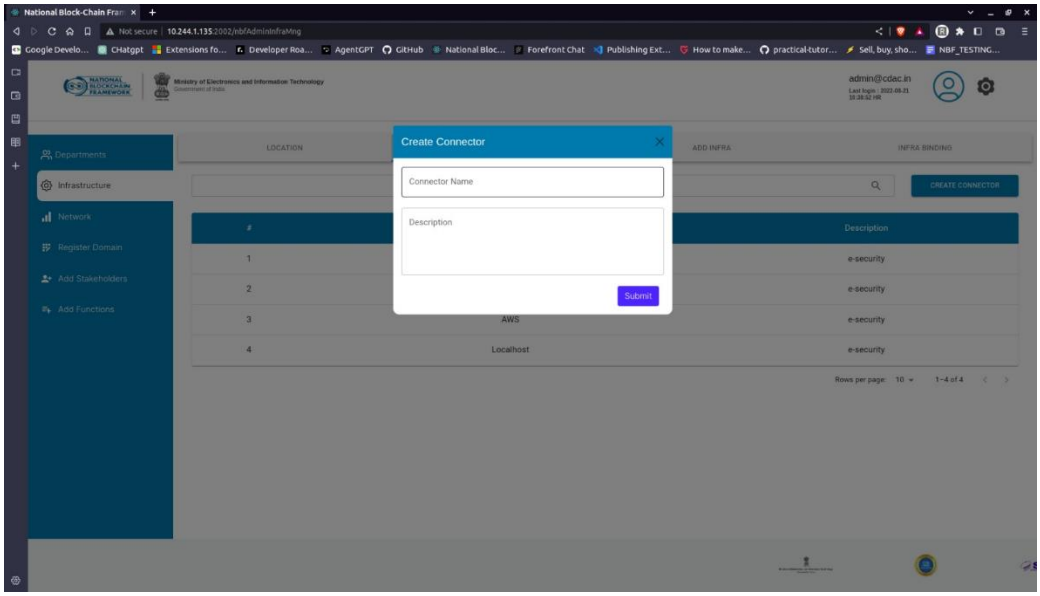


4.2.2. Connectors

The type of connectors supported by BaaS are available through connectors tab under Infrastructure Menu. There is an option for Generic and AWS options.

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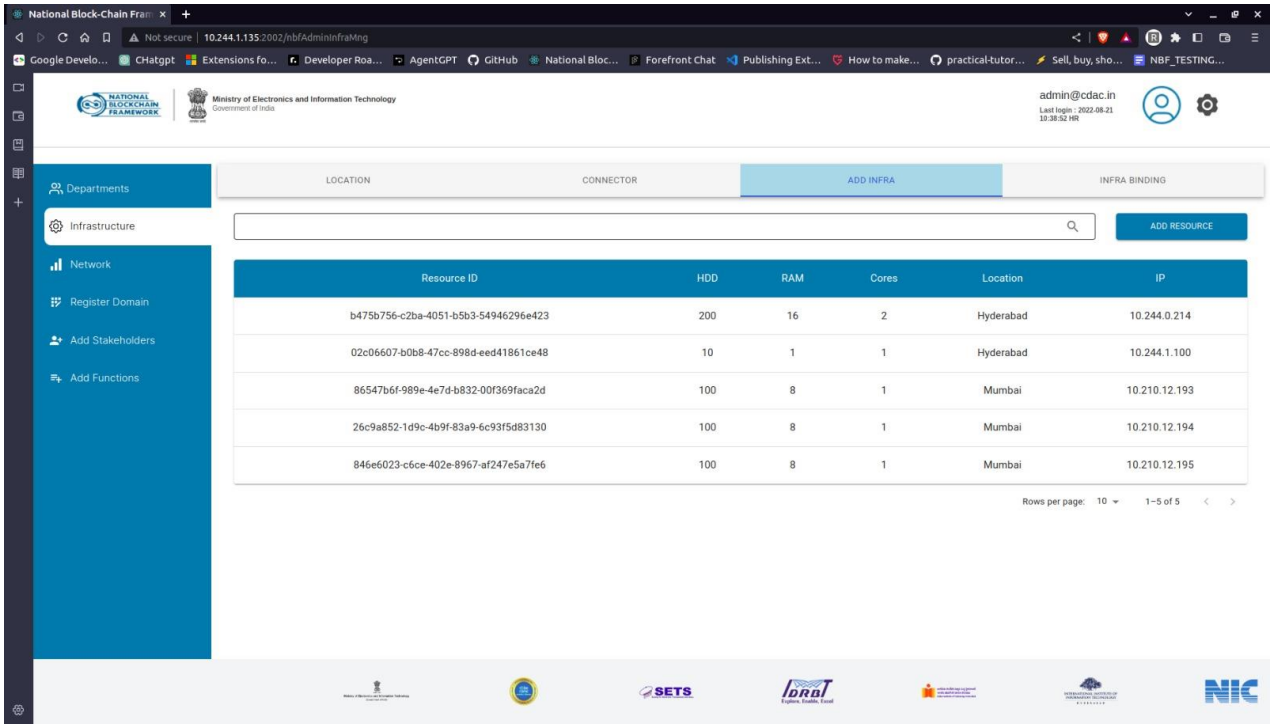


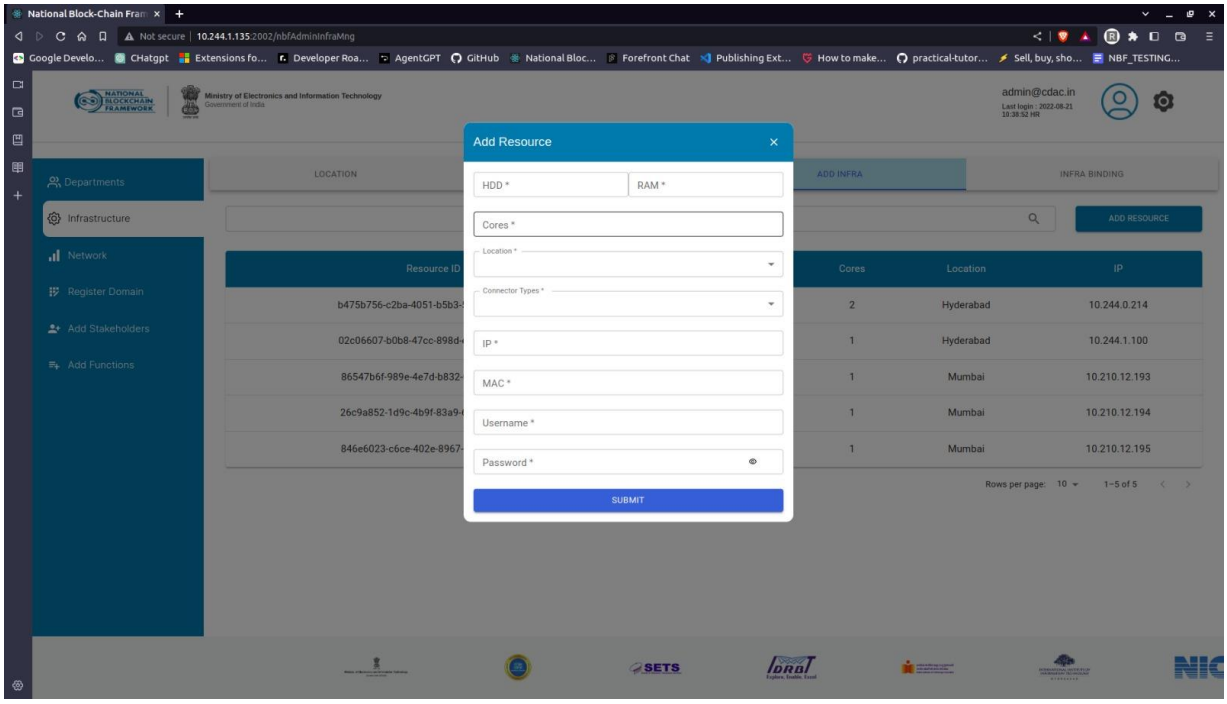
4.2.3. Add Infrastructure (Generic Connector)

Admin should add the resources and make them ready for mapping them to the resource requests received from various departments

Under the Infrastructure menu item, click Add Resource button and provide the necessary details and submit it. **Please ensure that the resource should be reachable and username and password are to be provided mandatorily for automating the process of provisioning the resources**

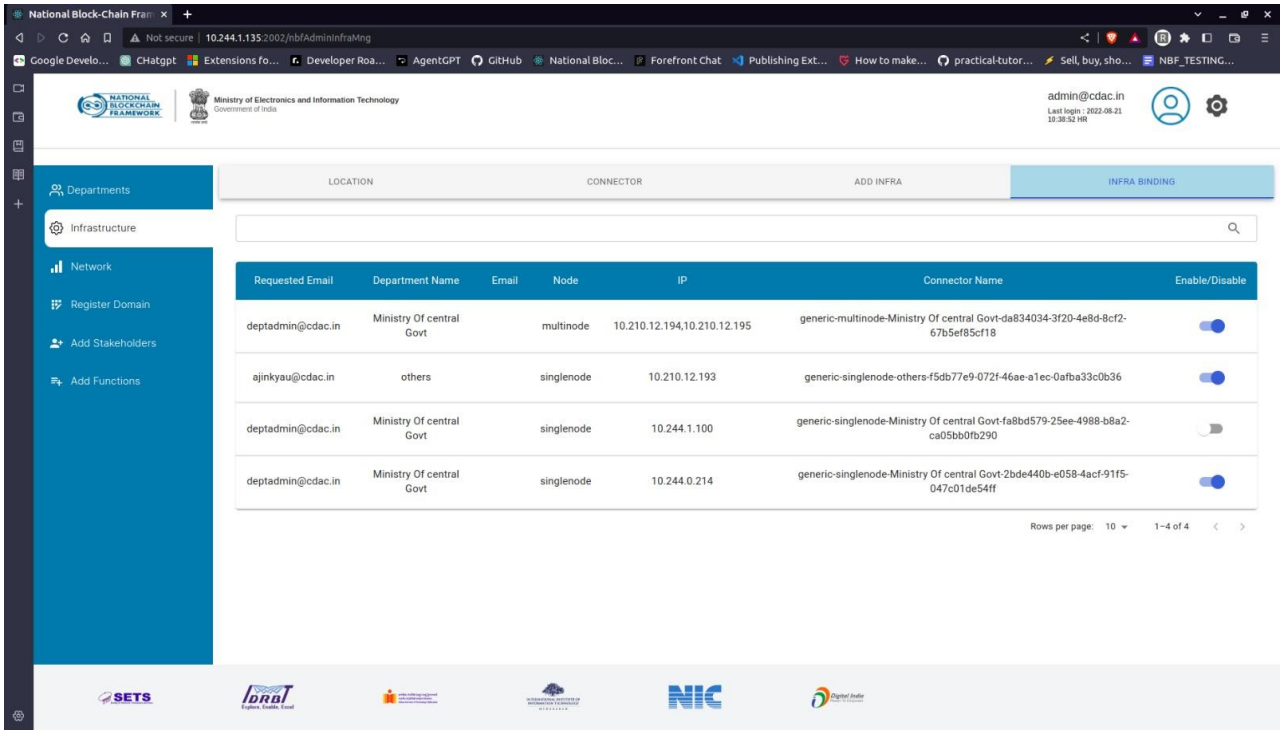
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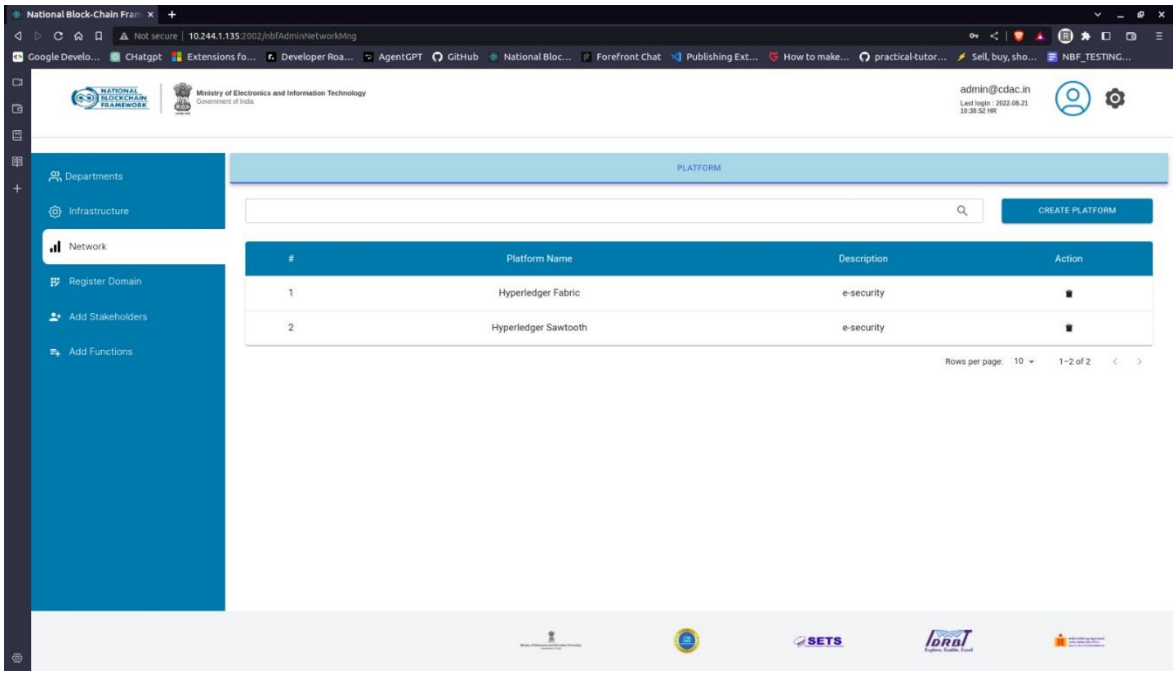
4.2.4. Approve Infrastructure request (Generic Connector)

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4.3. Network

The platforms supported by NBF BaaS are listed under this menu. Currently, it supports Hyperledger Fabric and Sawtooth platforms



5. Role: NBF Dept Admin role

Department admin can do the following:

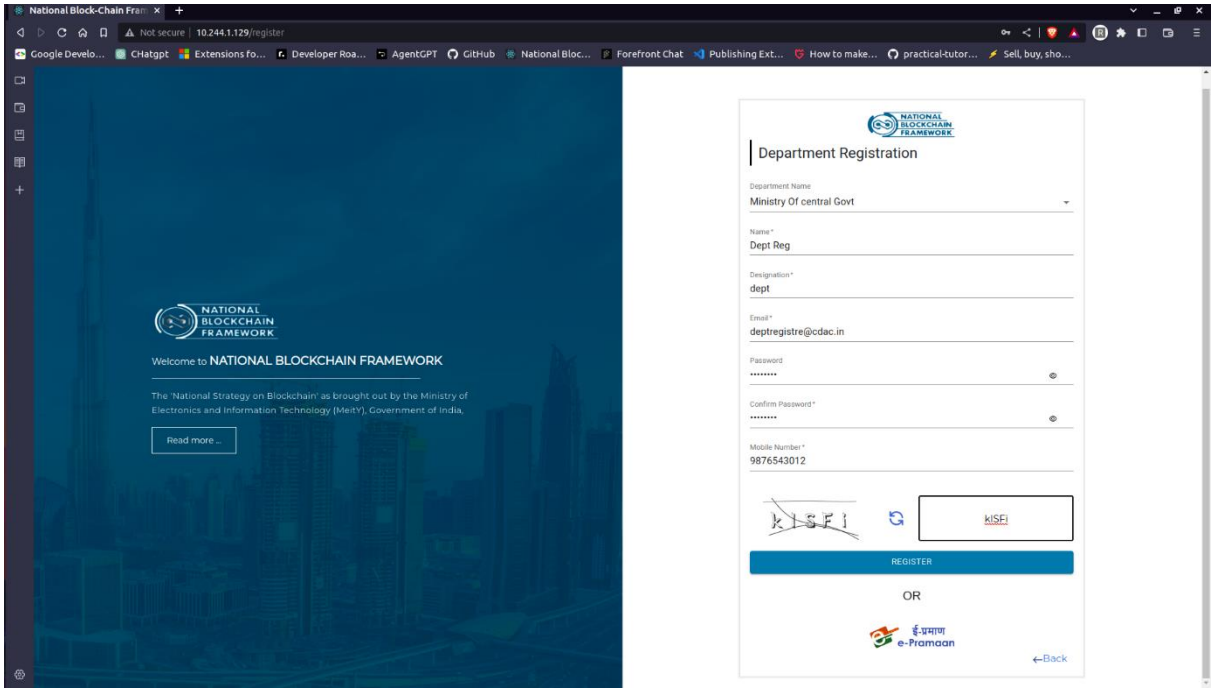
a. Register and Login

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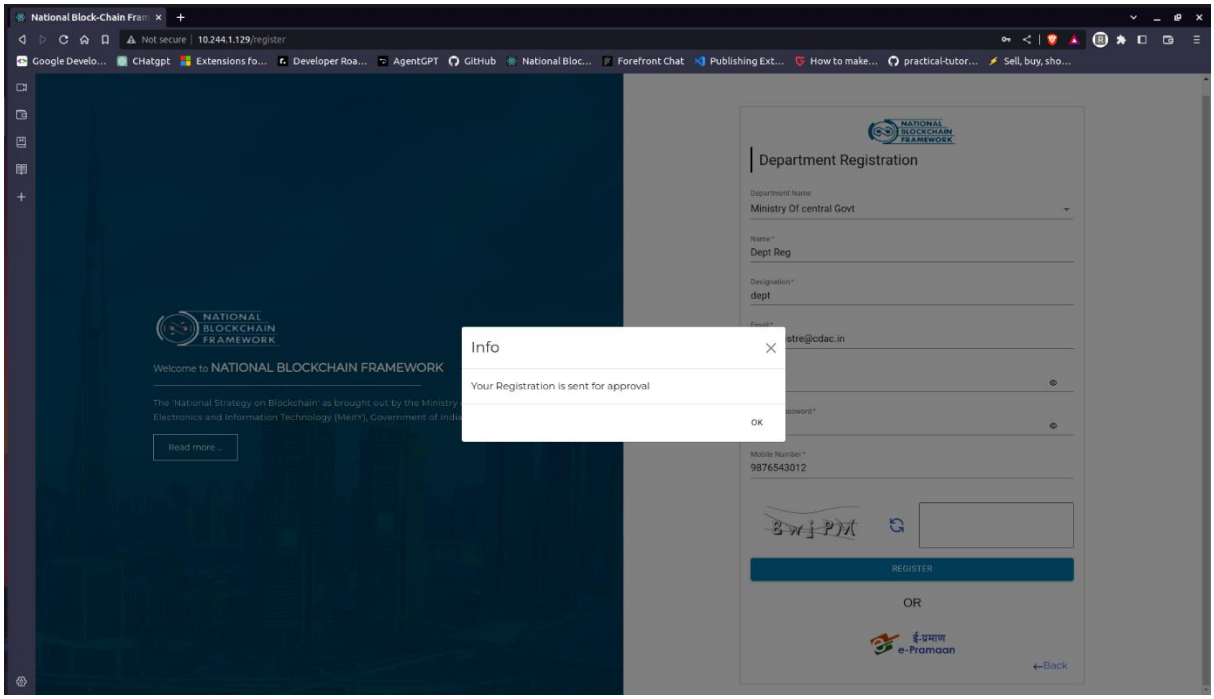
b. User Management: Creates NBF App Admins and enables them
c. Request for Infrastructure – Generic Connector
d. Create the Infrastructure for NIC and AWS connector types
e. Binding Infrastructure with a NBF App Admin
f. Application
    1. Configure the application
    2. Blockchain stack creation
        i. Network setup - includes channel creation, Choosing the option of file storage also
        ii. Deploy Chaincode - Deploy, Upgrade functions are available
        iii. User enrollment:
            1. creates the users - wallets are created per user
            2. view registered users for a particular application

5.1. Register and Login

- Register to the framework by submitting the required details

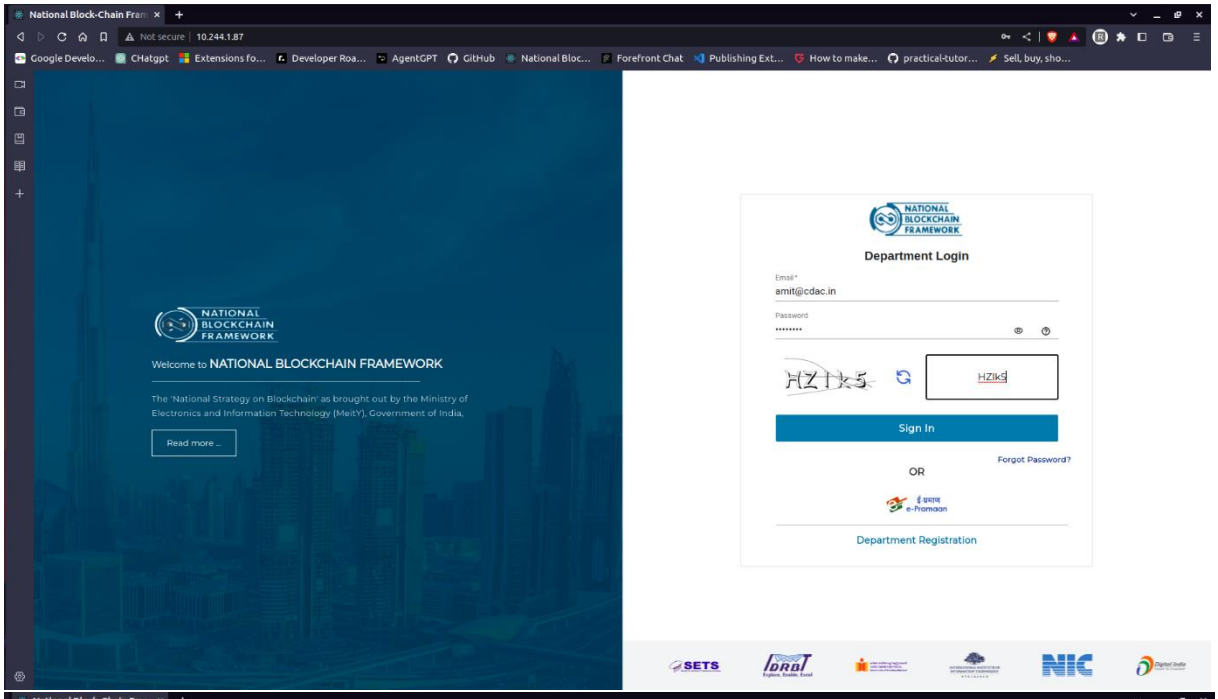


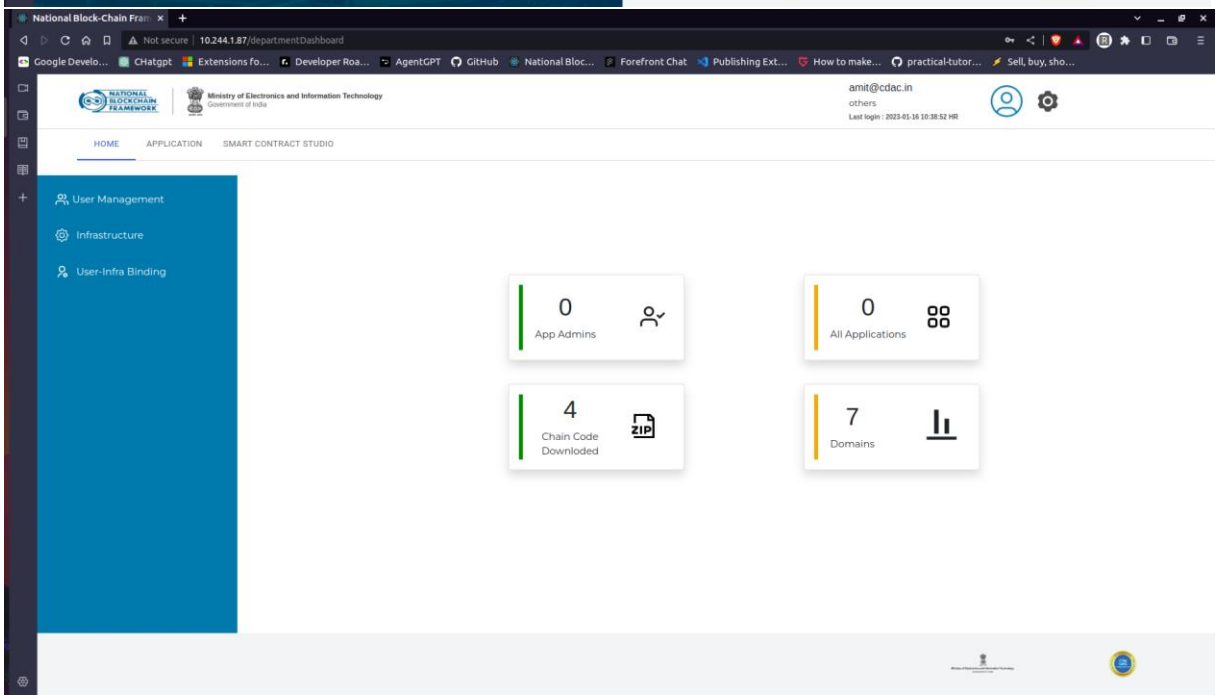
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- <NBF Admin will approve>
  After NBF Admin approves,
- Login with the credentials

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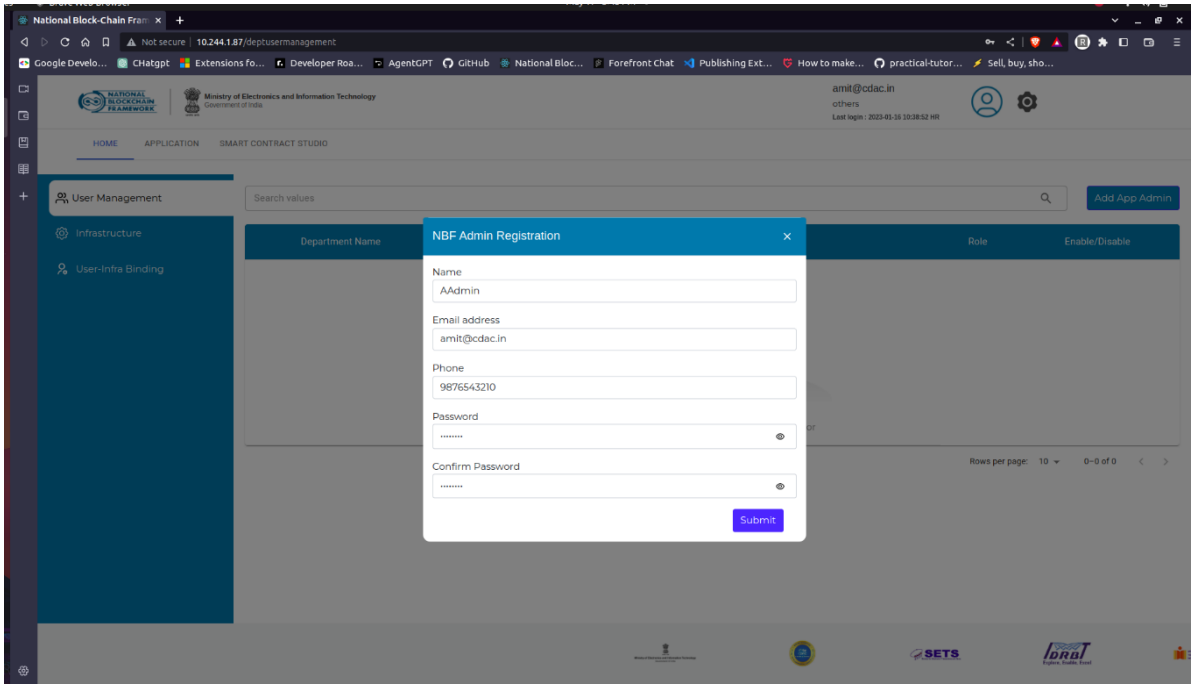


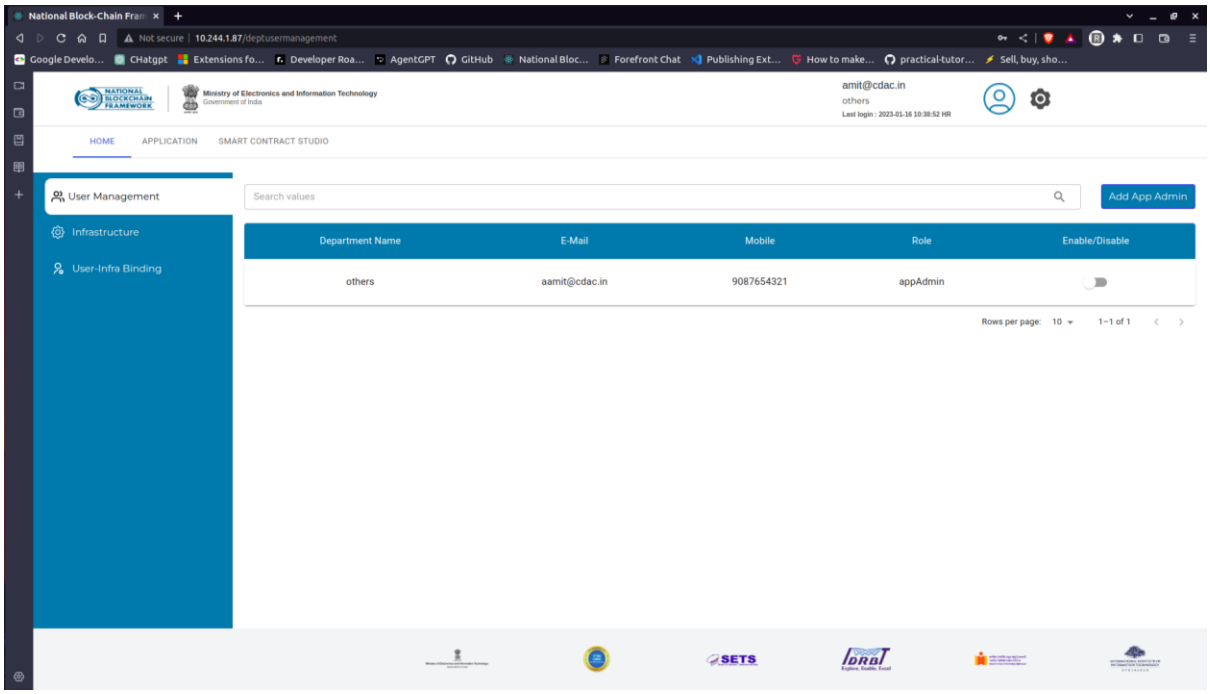
5.2. User Management

After login, Department admin can manage multiple App admins for different applications. He can add App Admins by providing minimum details and enable it. Using these credentials, App Admins can login.

Provide the details of NBF App Admin

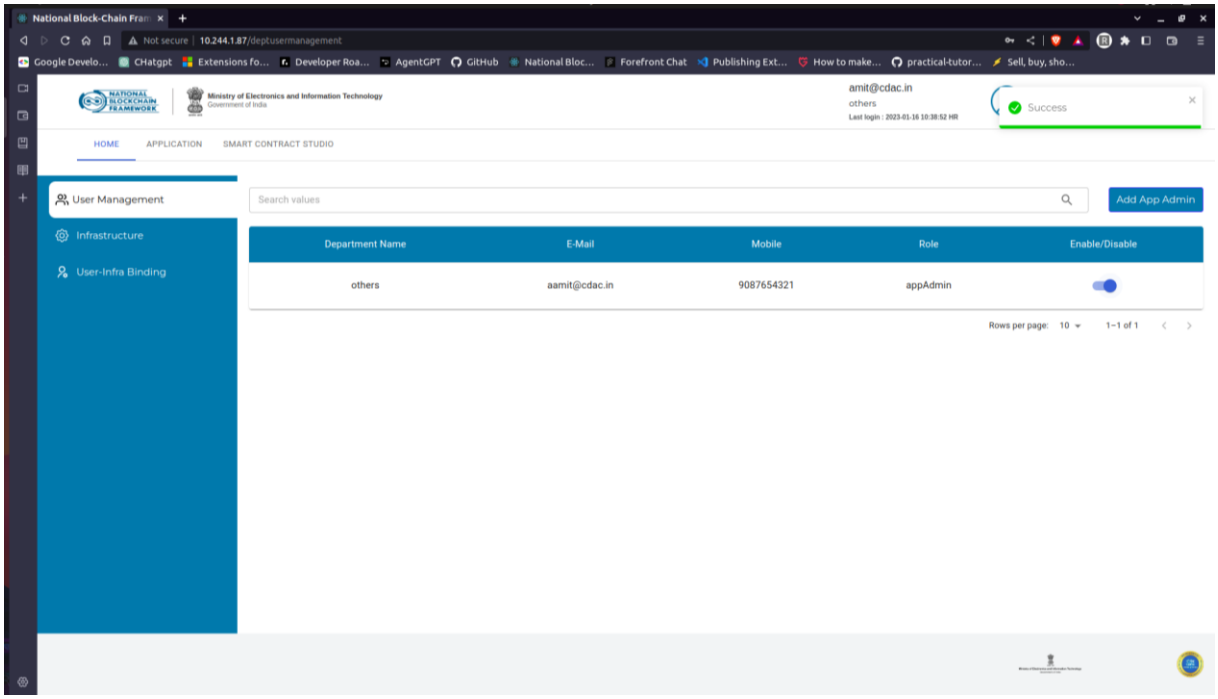
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Enable the App admin for login

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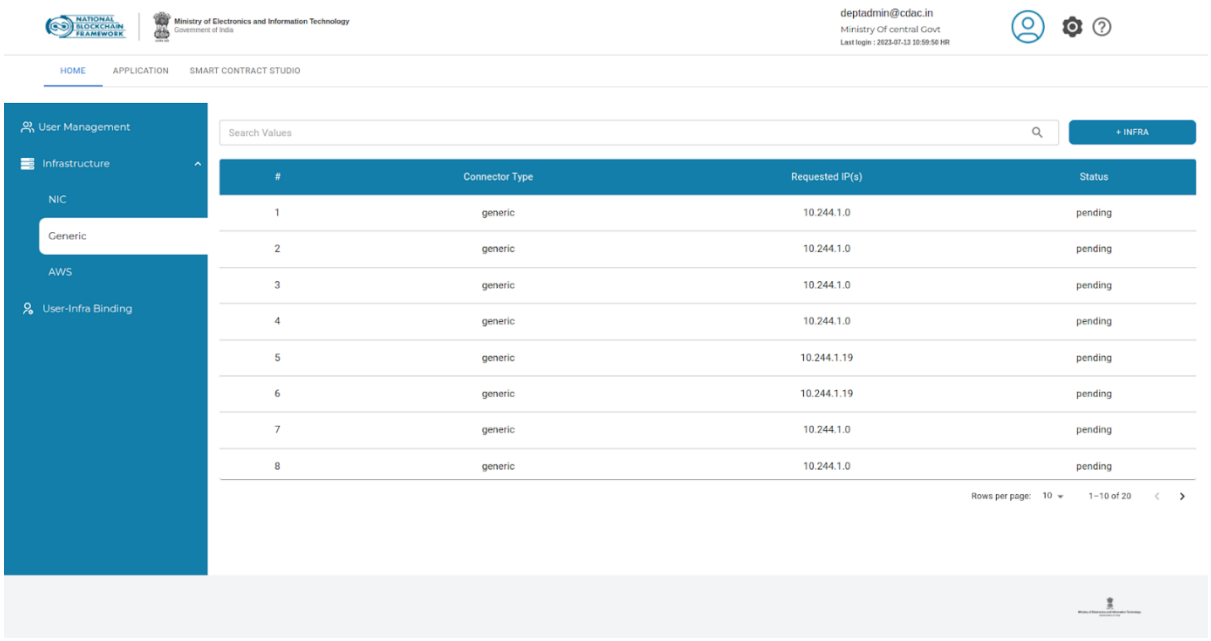


5.3. Infrastructure

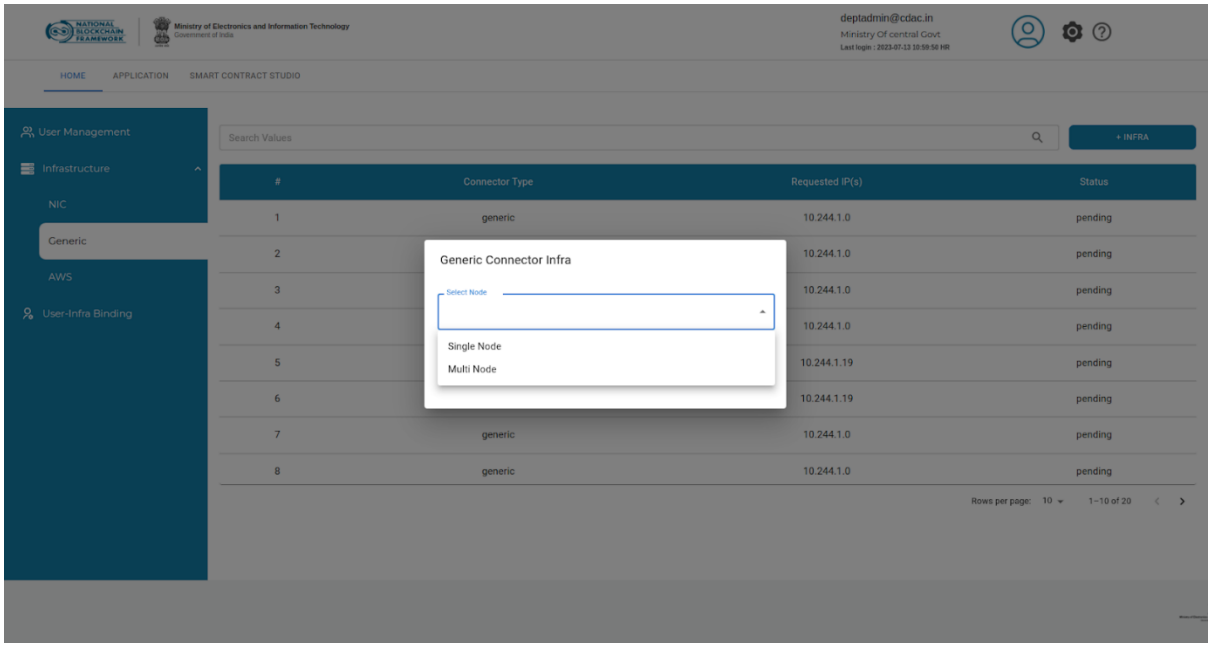
NBF BaaS supports Bring Your Own Infrastructure. Currently it supports Generic and AWS connectors.

5.3.1. Generic Connector

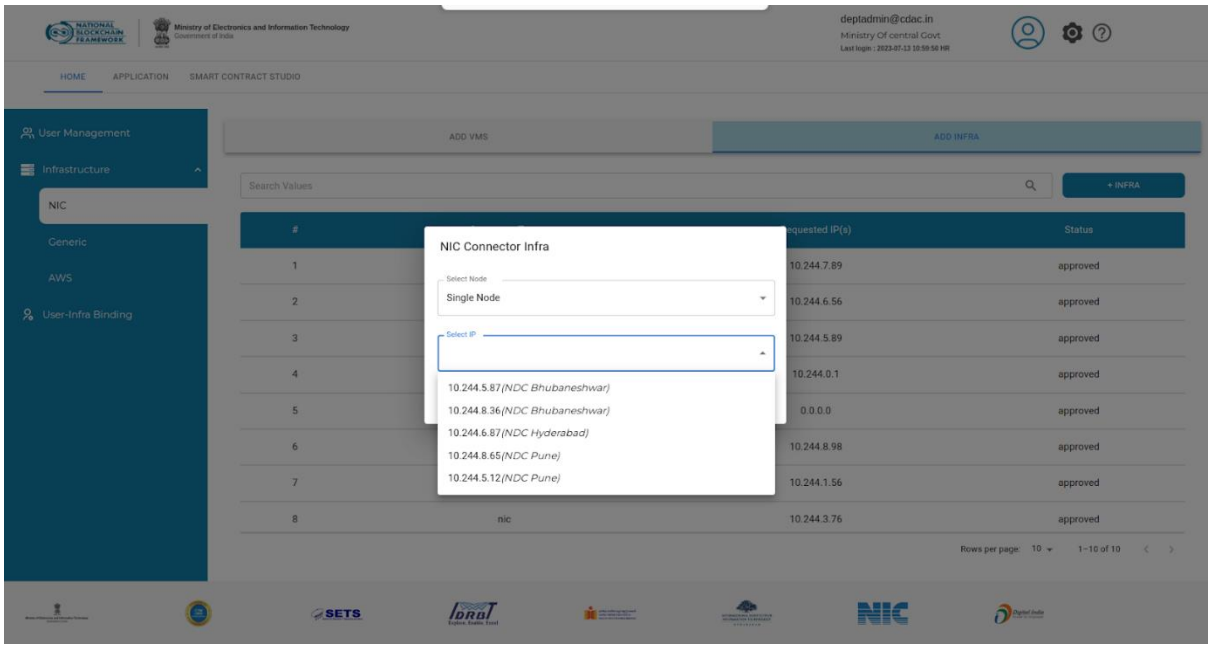
Department admin can request for the Infrastructure by specifying the details about the type of the blockchain application being planned. This includes, Specifying the infrastructure type, type of deployment (single node), followed by selecting the VMs.



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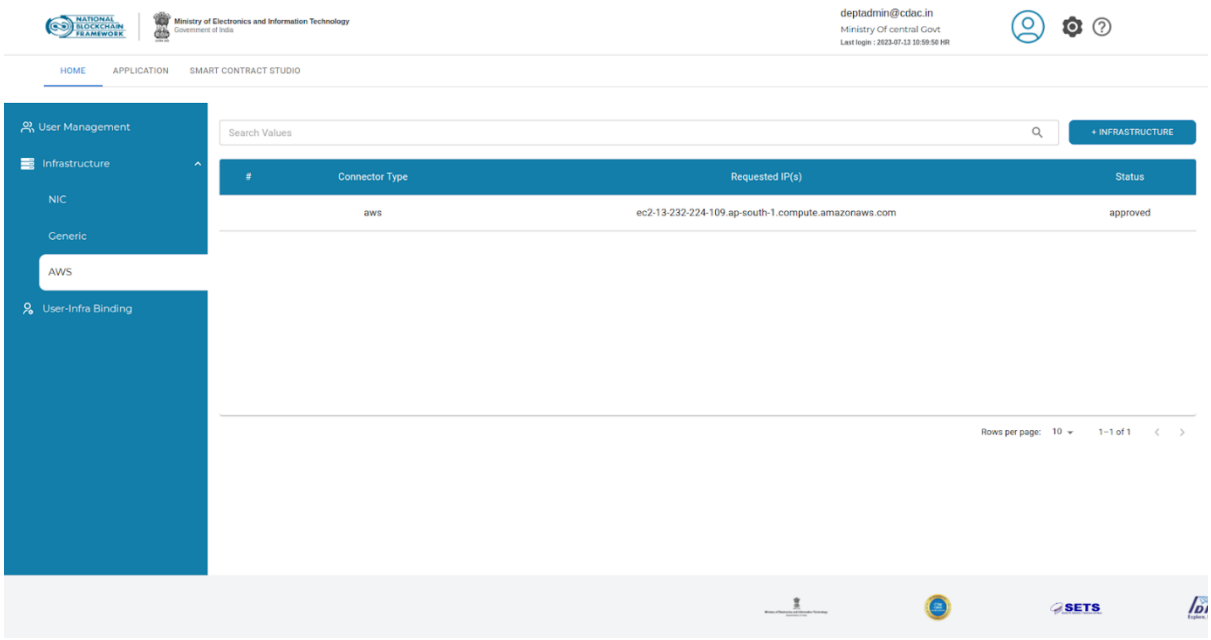
The request will be processed by NBF admin. Till that time, it will be in a pending state.

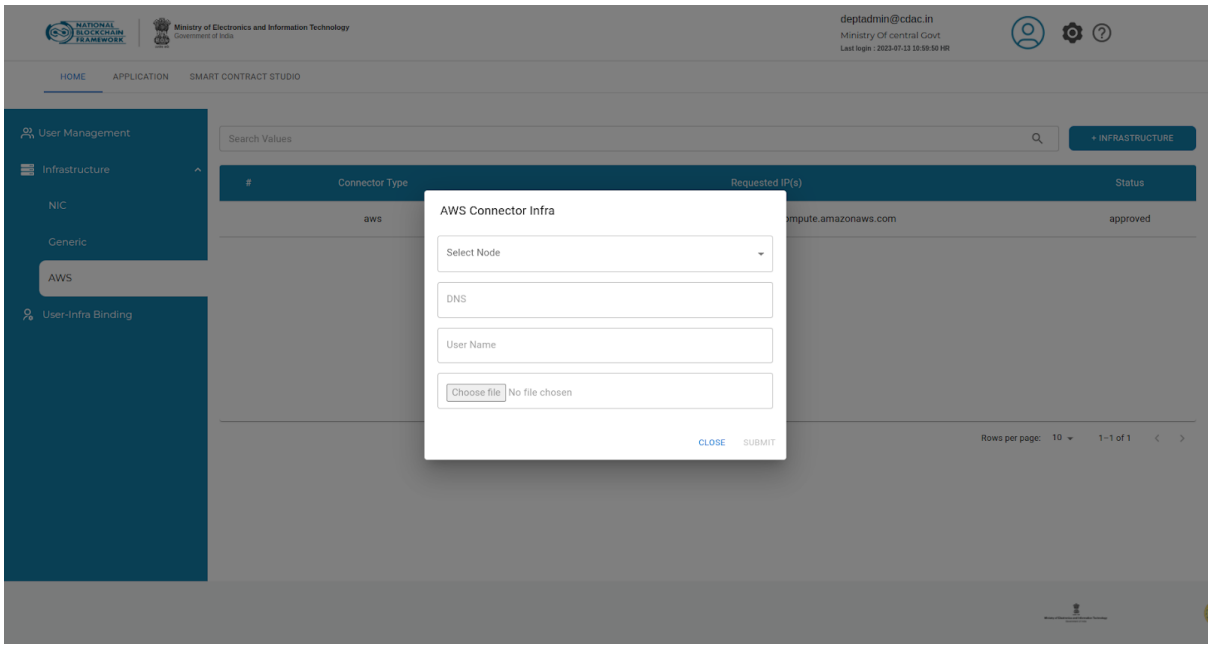


5.3.2. AWS Connector

Provide the necessary details required to connect to the resources in the AWS cloud and continue with the network setup (Currently this supports single node network setup)

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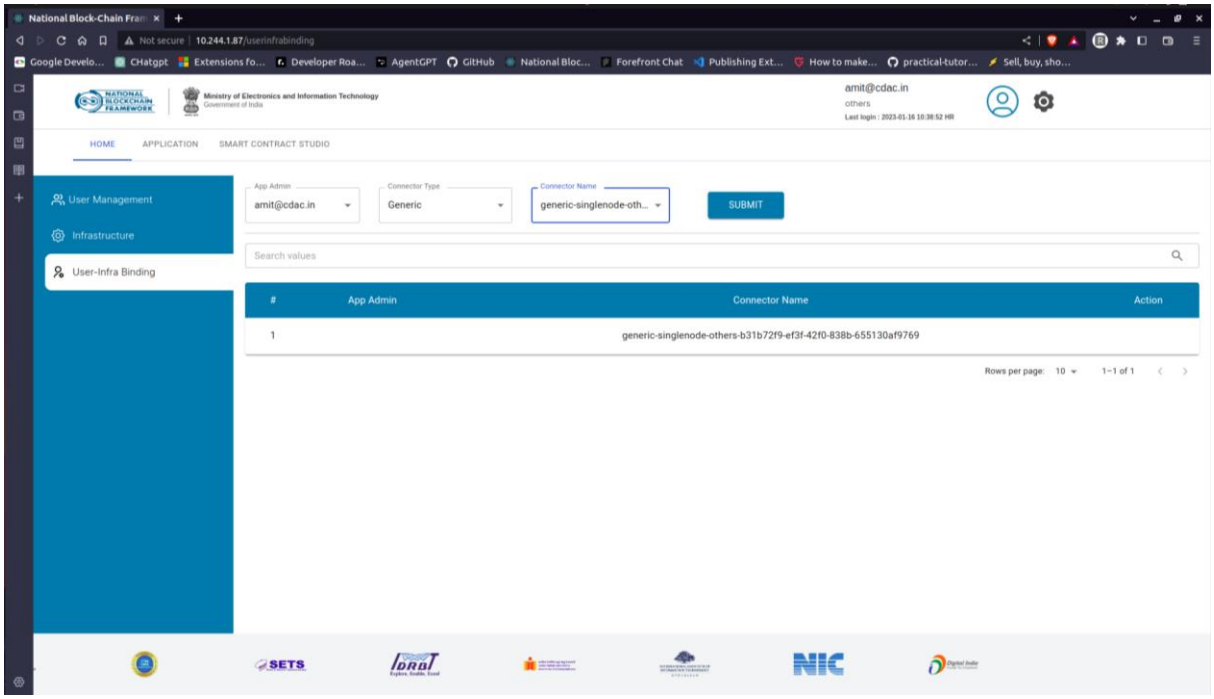


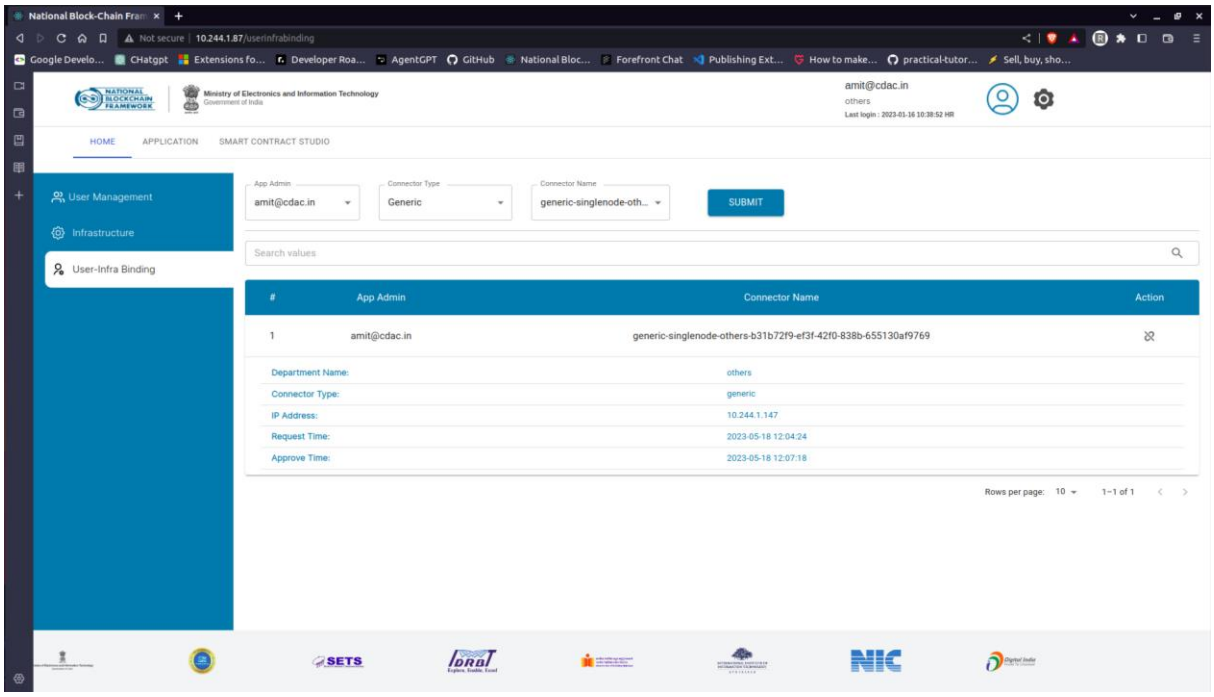
5.4. User-Infra Binding

After the infrastructure is approved by the Admin, Department admin can bind that infrastructure with a specific NBF App Admin or to Self.

Choose the AppAdmin, Connector type and the connector name and submit the details for user – infra binding.

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Once the binding is completed. Application setup can be initiated either by Department admin or App Admin (depending on the binding)

6. Role: NBF App Admin

He can login with the credentials provided by the department admin. He can setup the complete application with the assigned infrastructure.

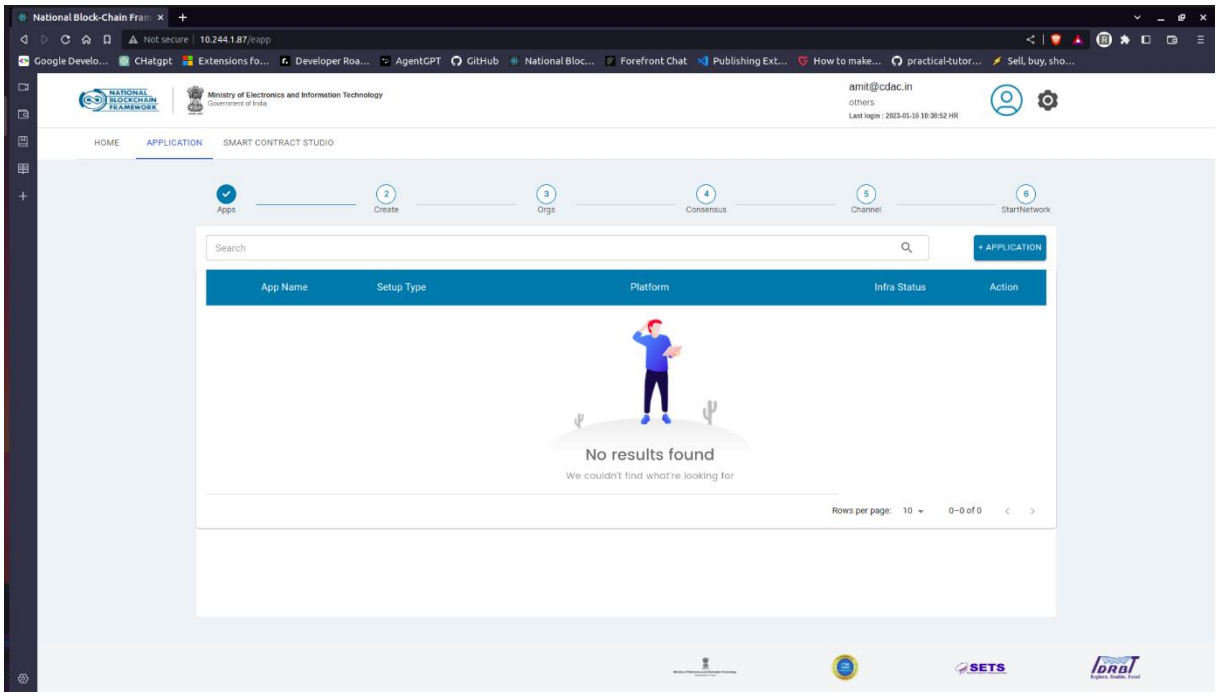
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7. Application

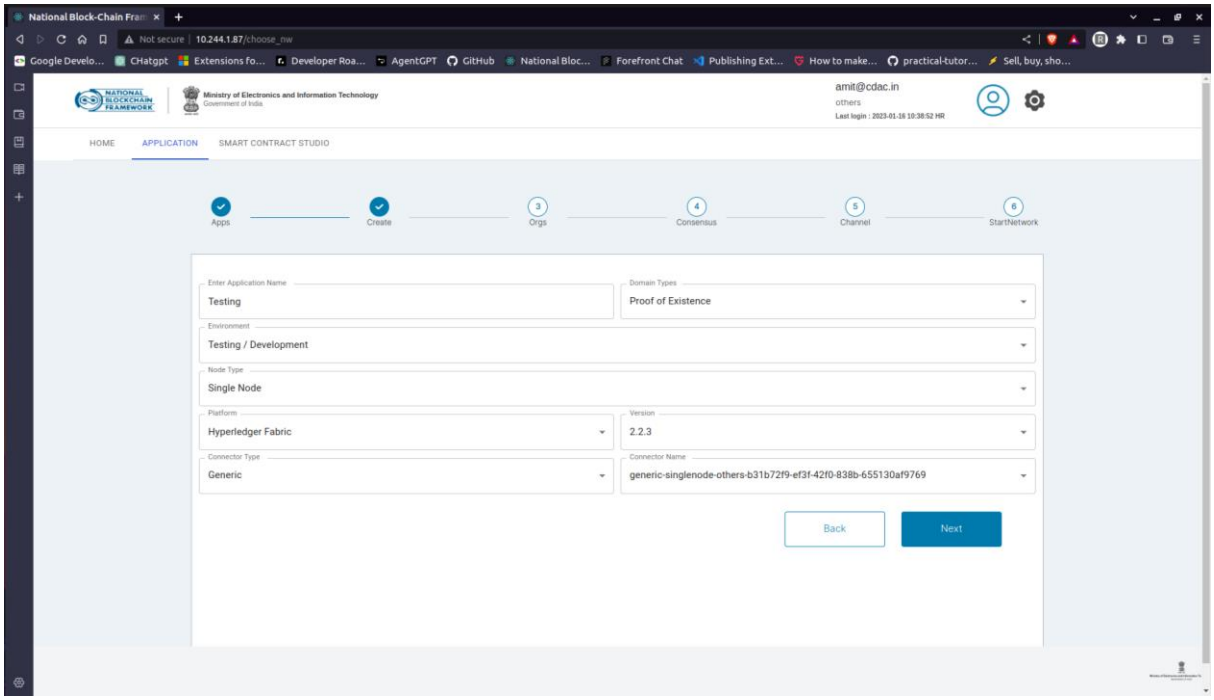
Department Admin or NBF App Admin can setup the BCT application through this.

7.1. Network setup

Click on **+Application** and provide the minimum details about the application.

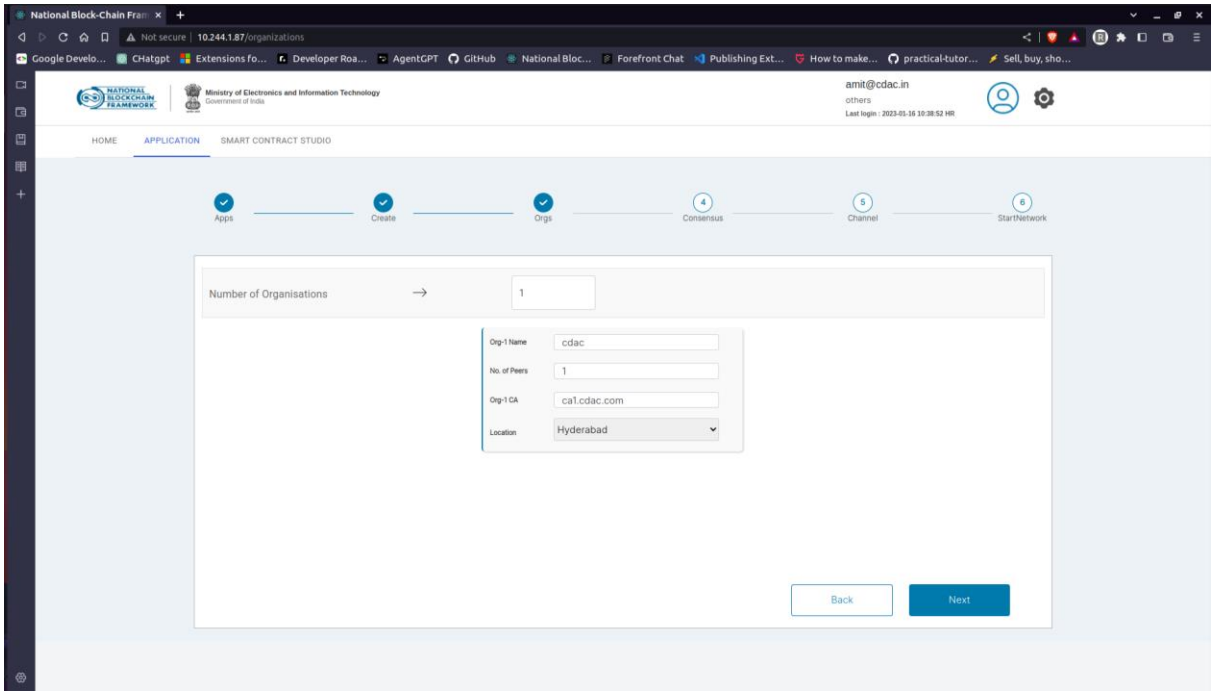


Application details:

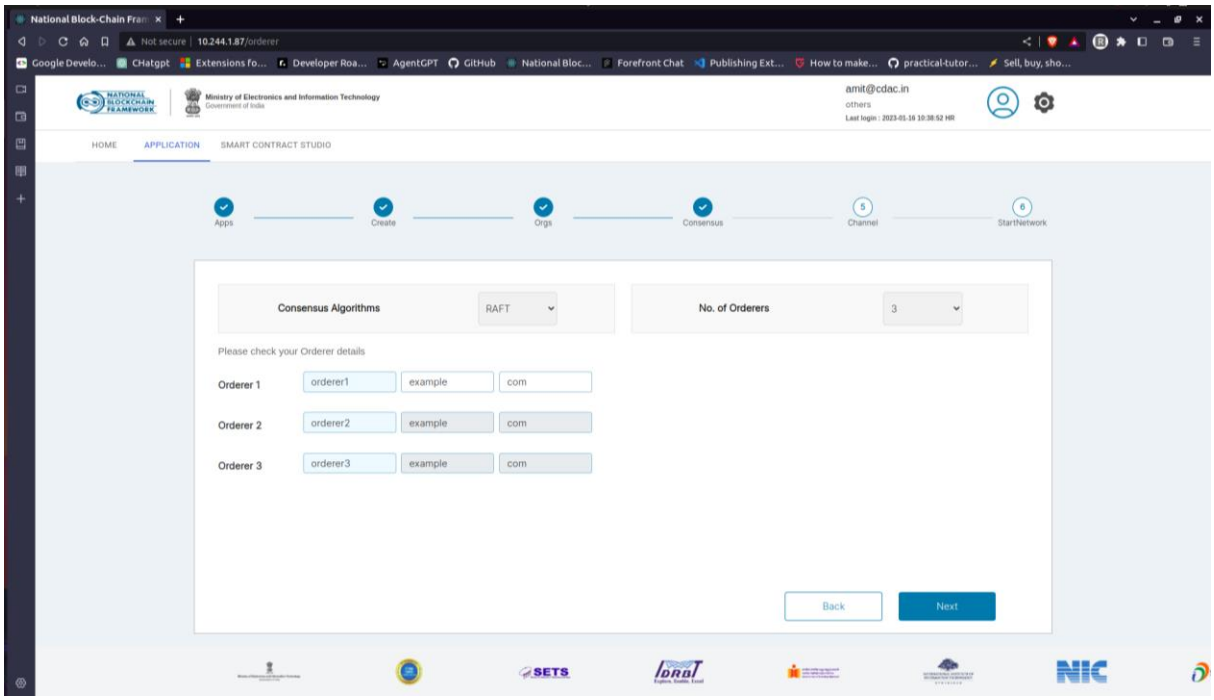


For Fabric Network setup, provide the organization details

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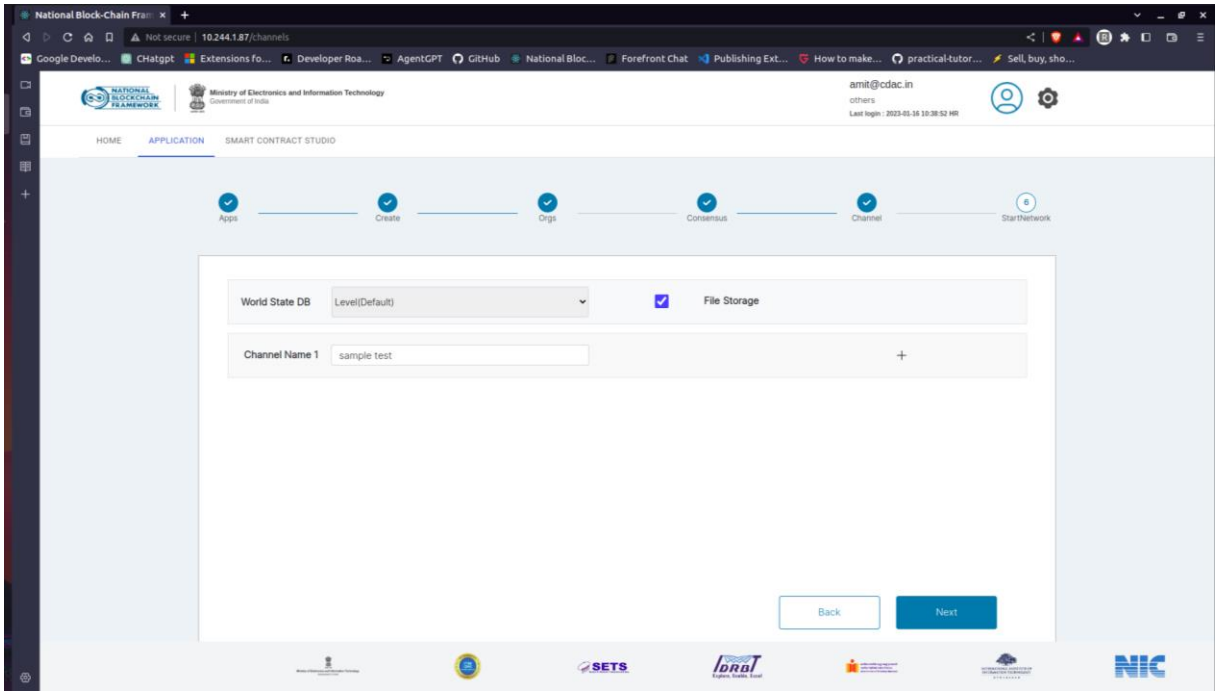


For Fabric Network setup, provide the number of orderers

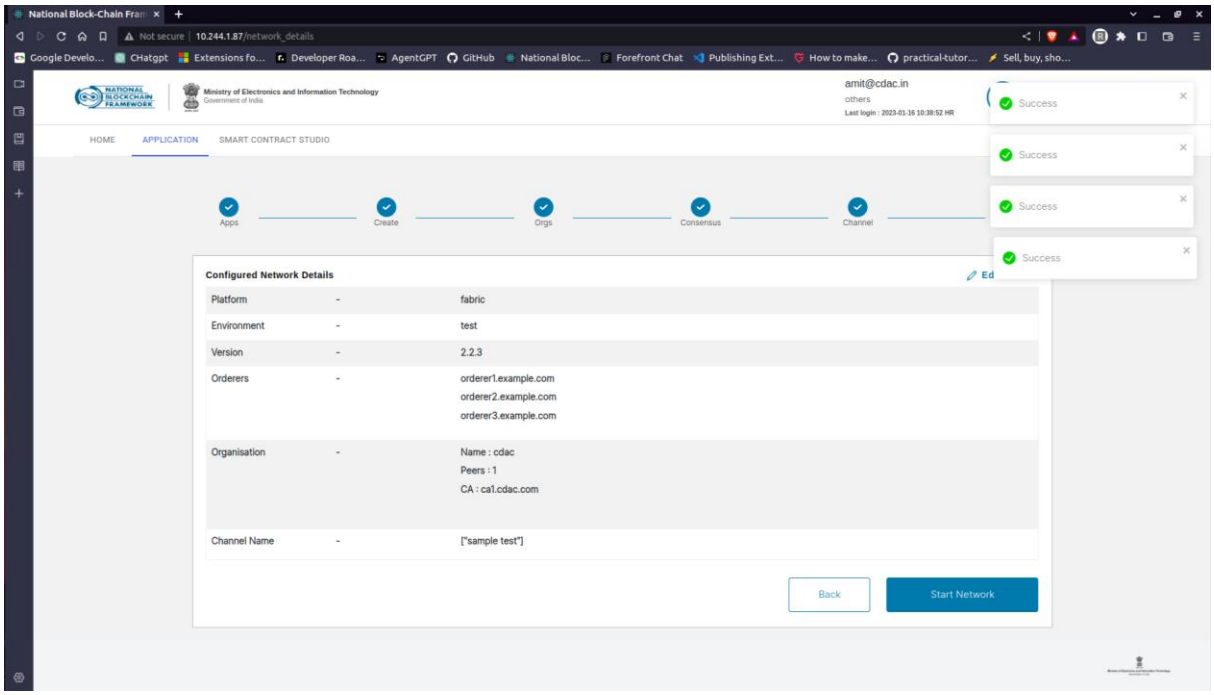


For Fabric Network setup, provide the world state DB details, whether file storage is required or not and the channel name

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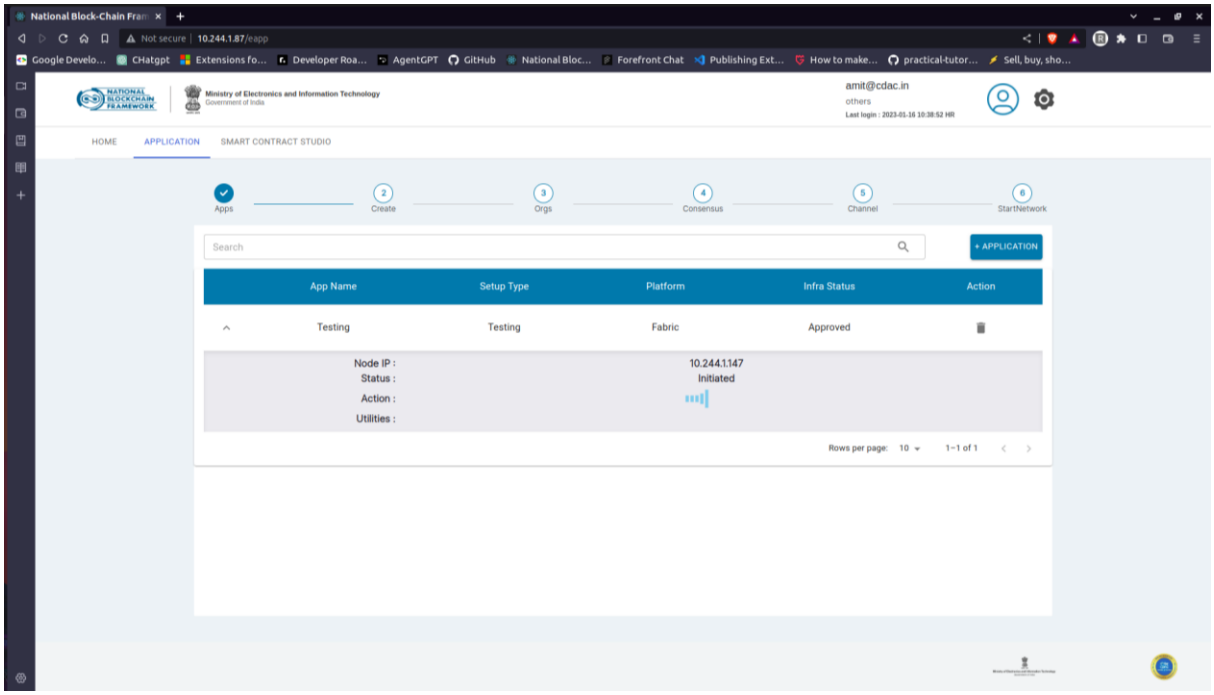


Verify the configuration details and submit for network step up

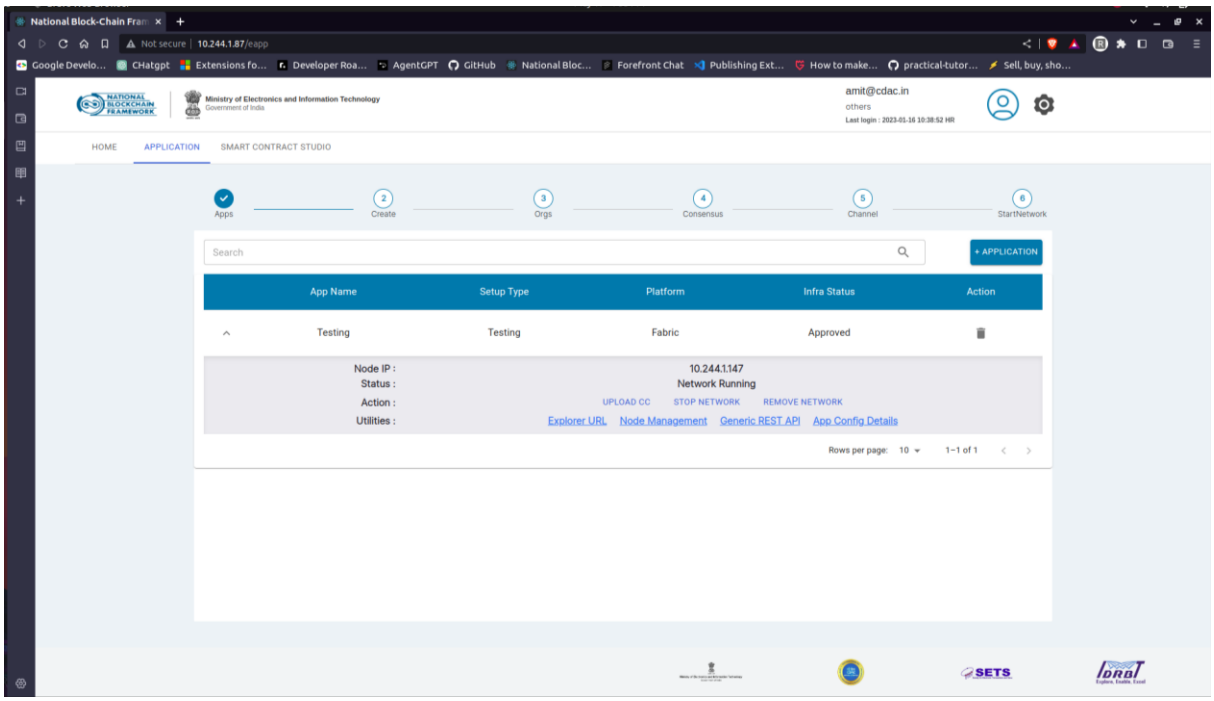


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Network setup is in progress

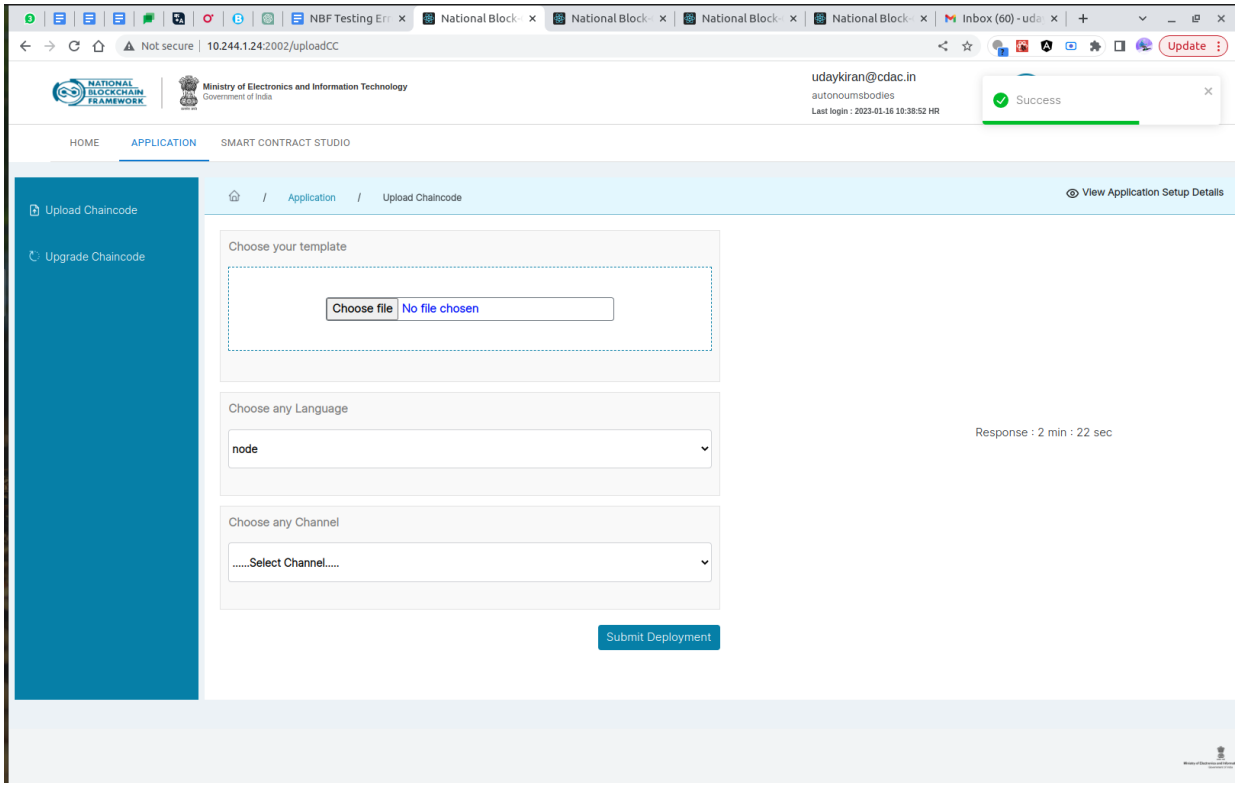


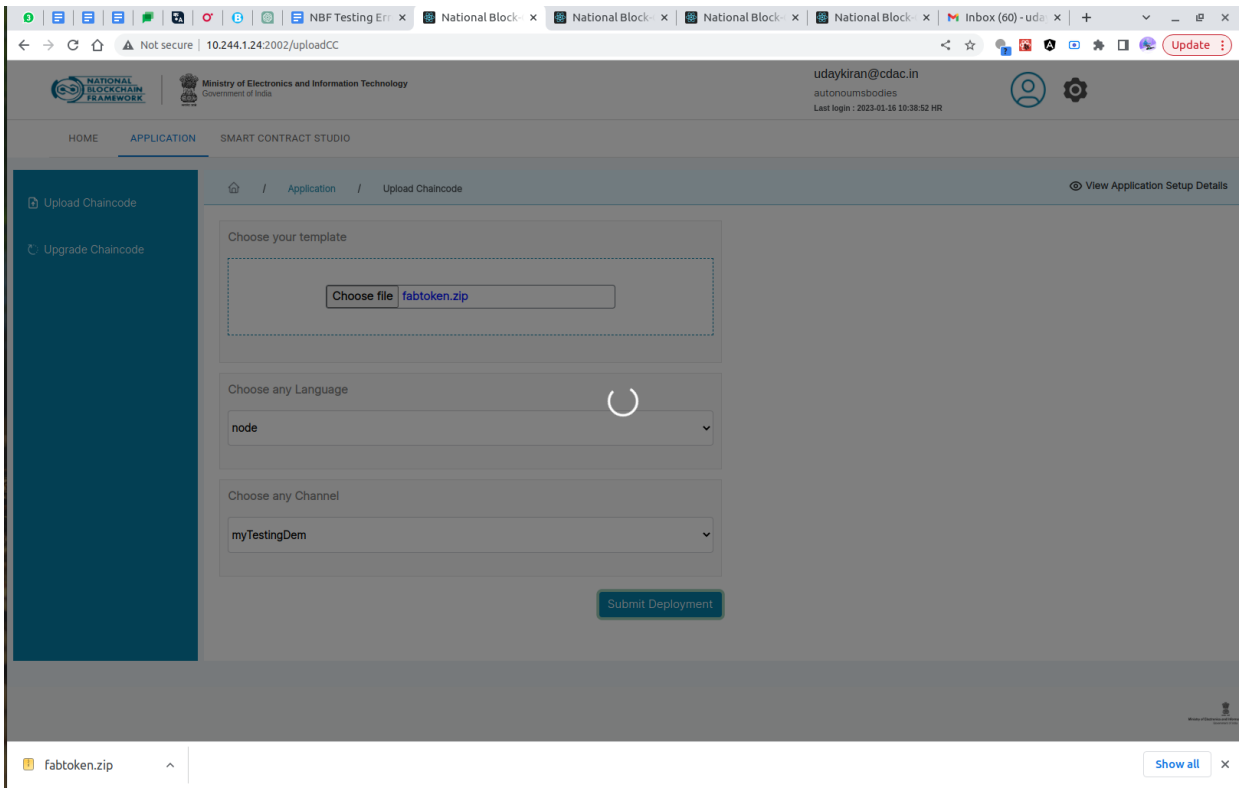
For Fabric Network setup, after the network setup, deploy the smart contract



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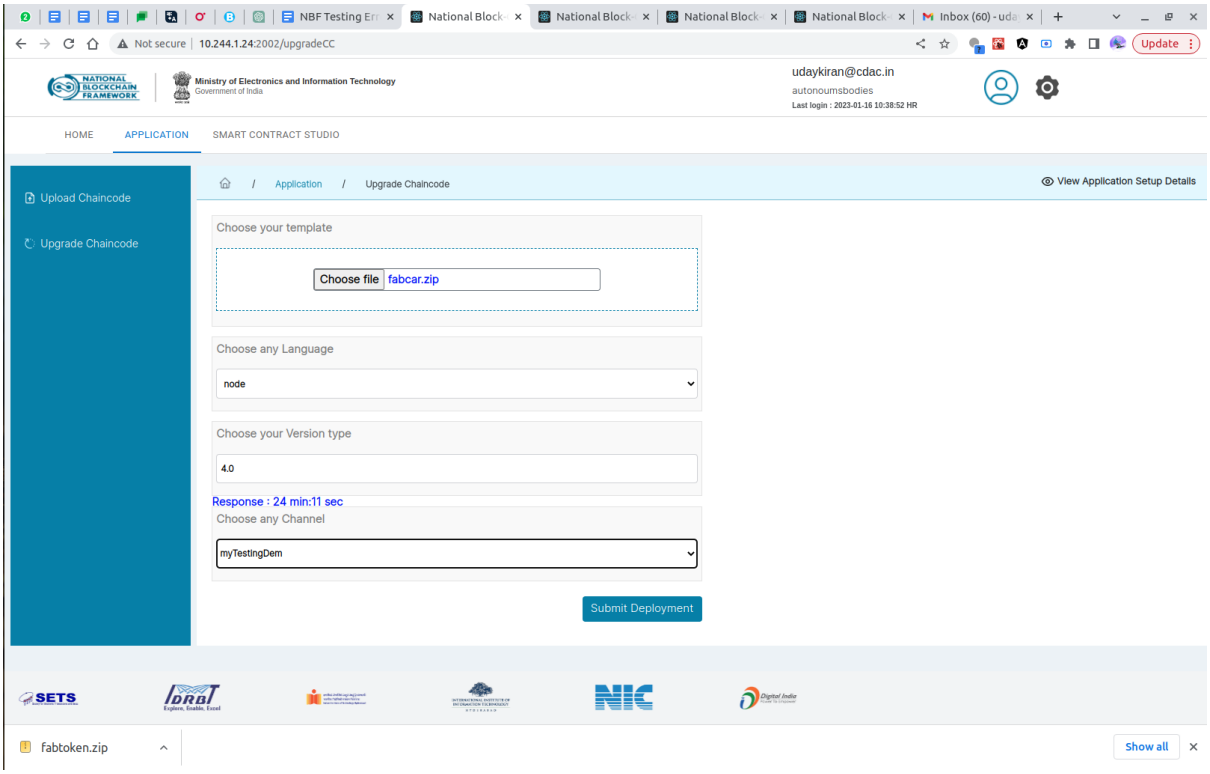
7.2. Deploy smart contract

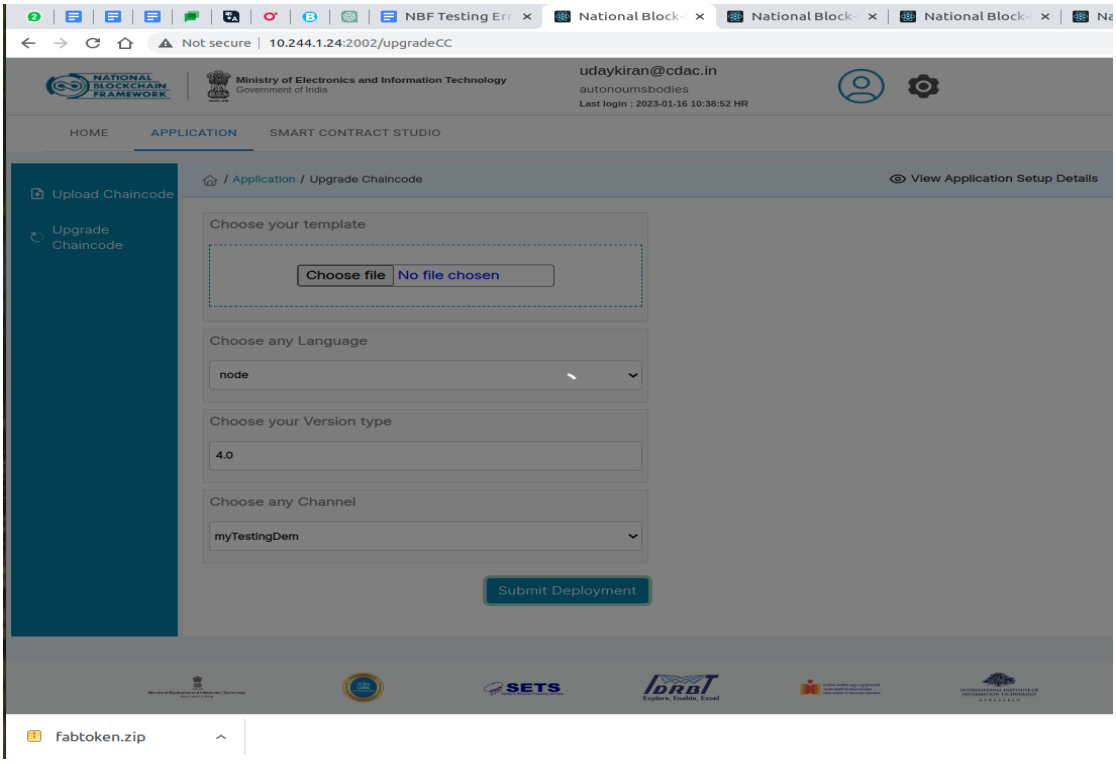




Smart contract upgrade:

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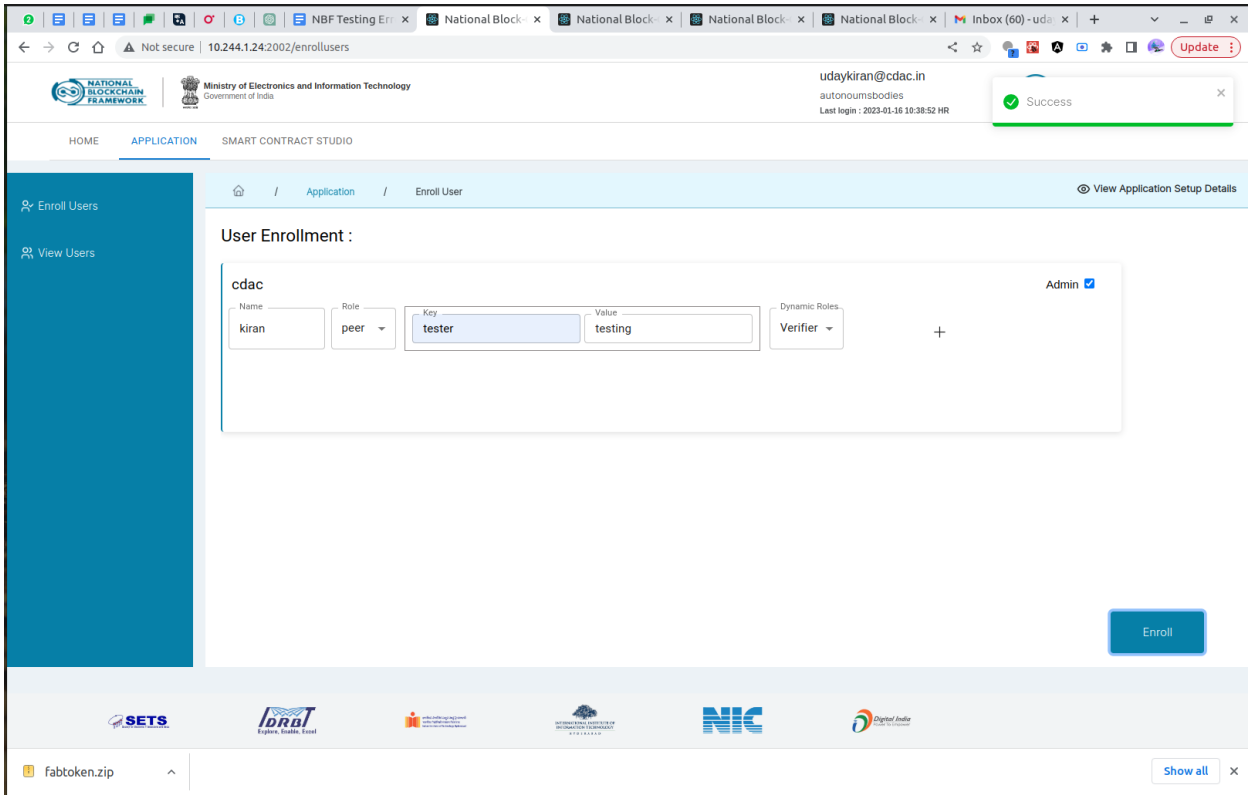


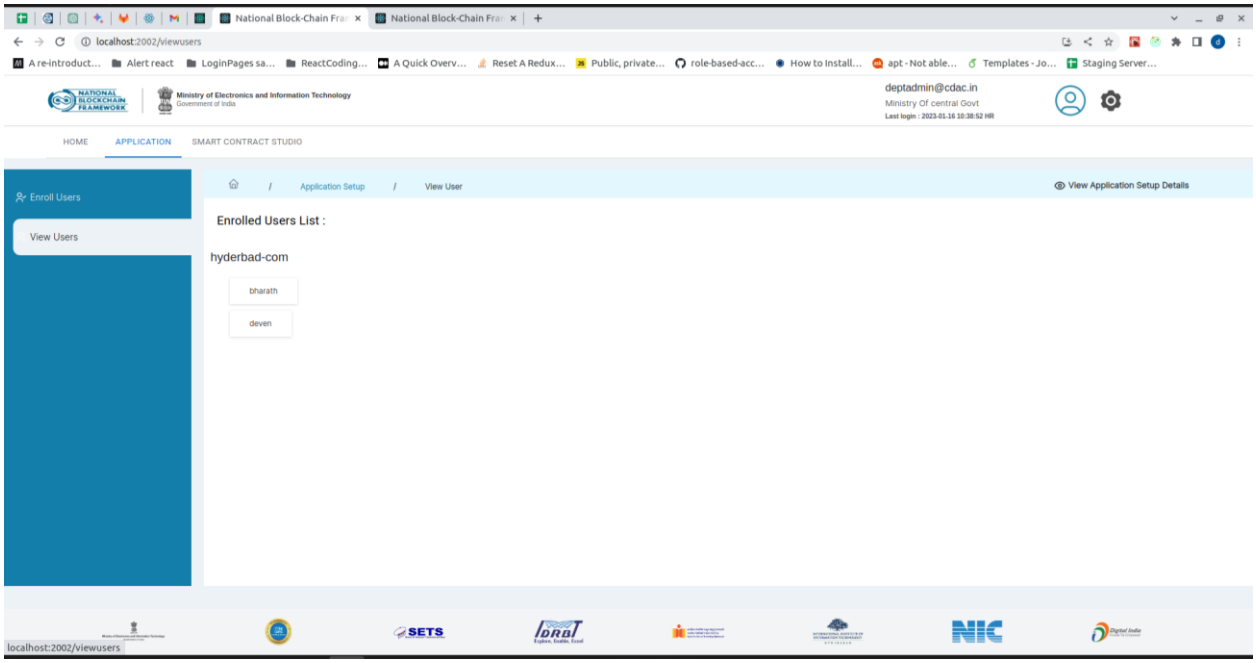


7.3. User enrolment

For Fabric Network setup, user enrolment (wallets will be created)

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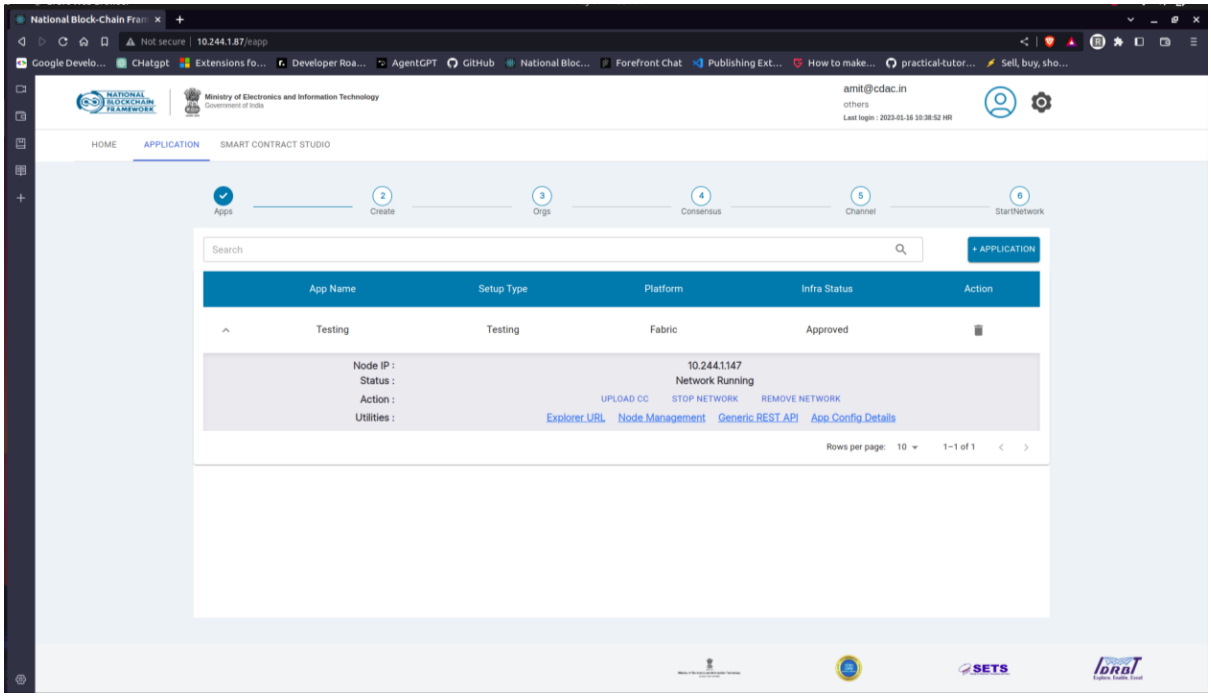


7.4. Application management

Explorer and Node management are provided per application

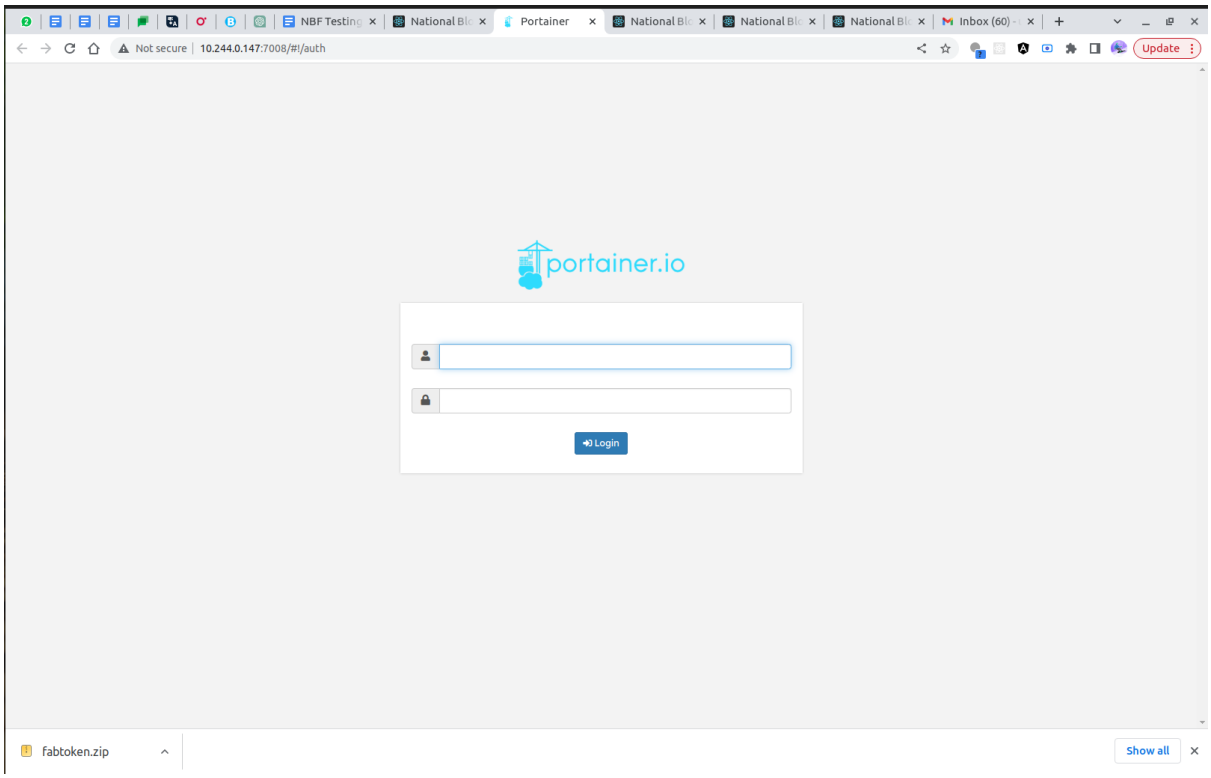
Configuration of the network for provided through **AppConfigDetails .** This has the details required for quick integration with the application. REST APIs for integration are provided through **Generic REST API.**

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7.4.1. Node management

Credentials are provided in **AppConfigDetails**

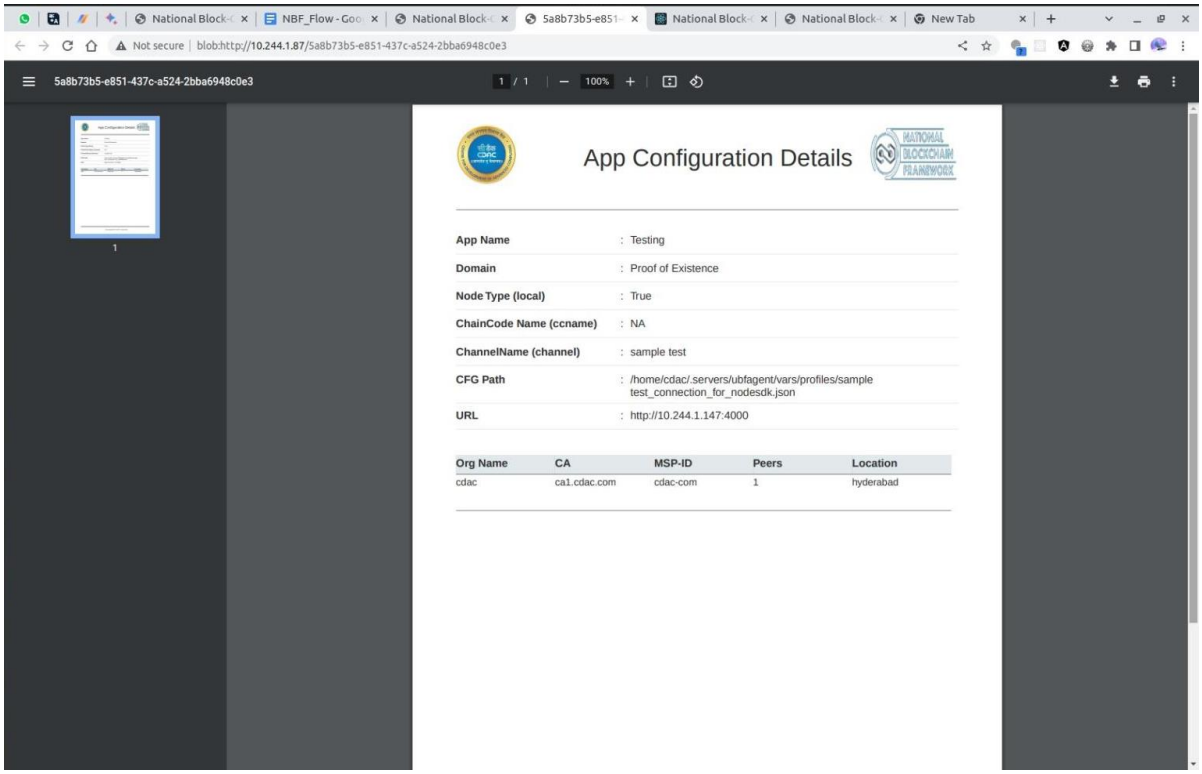


7.4.2. Explorer

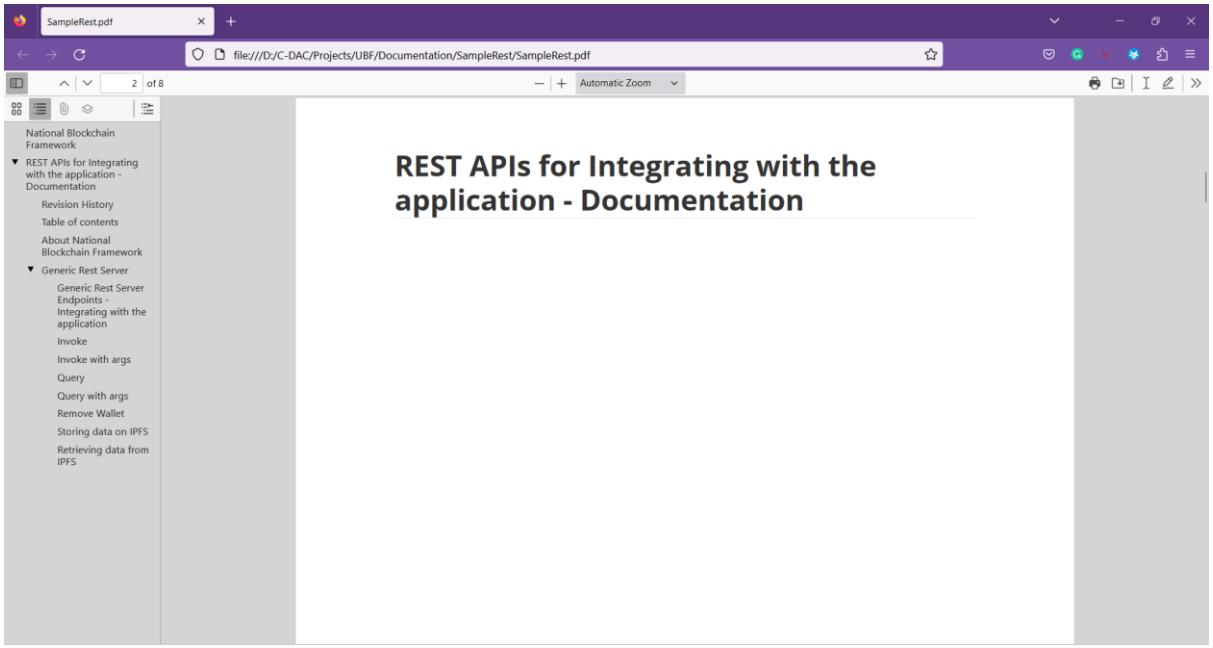
Credentials are provided in **AppConfigDetails**

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7.4.3. Configuration details



7.4.4. Generic REST APIs

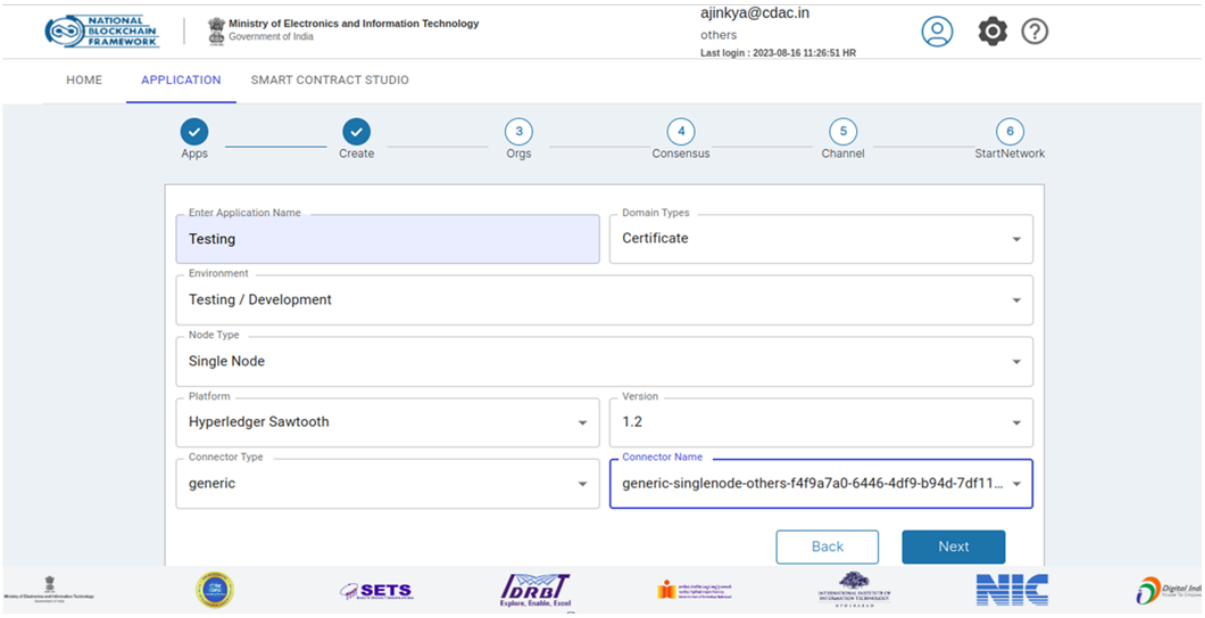


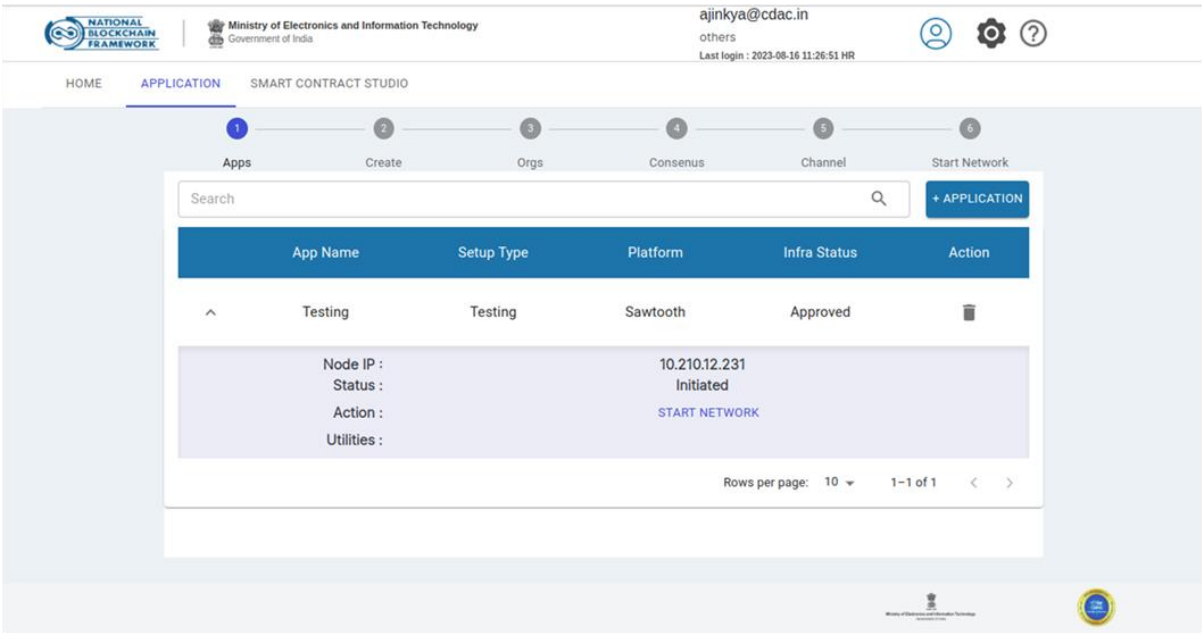
8. Application flow for Sawtooth

8.1. Network setup Sawtooth

Click on **+Application** and provide the minimum details about the application.

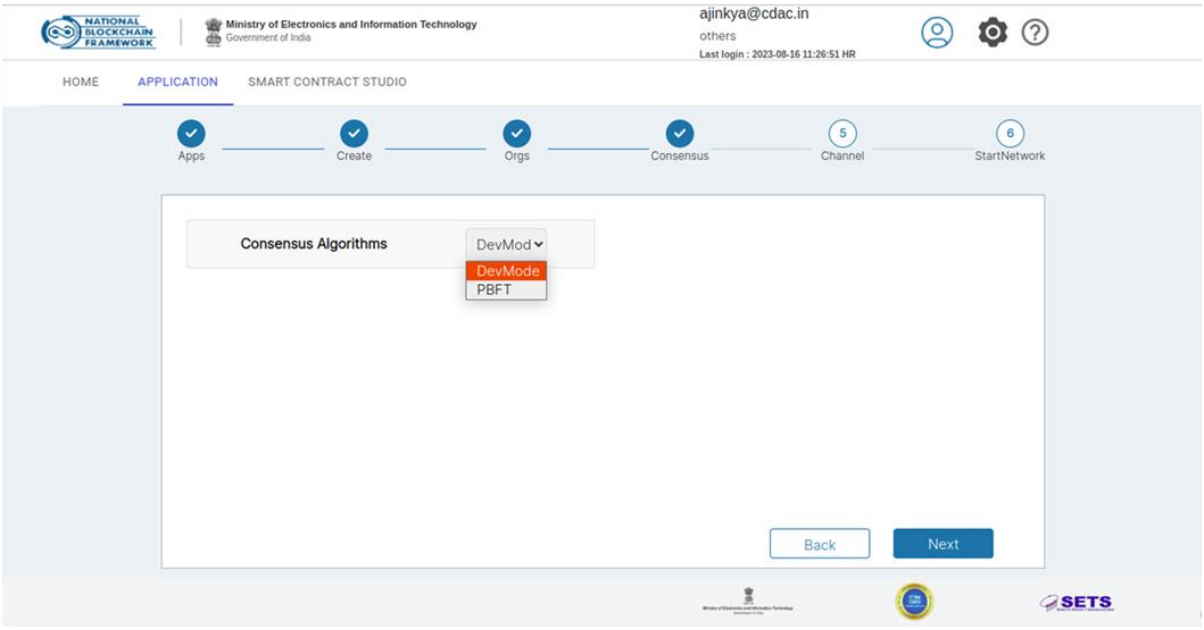
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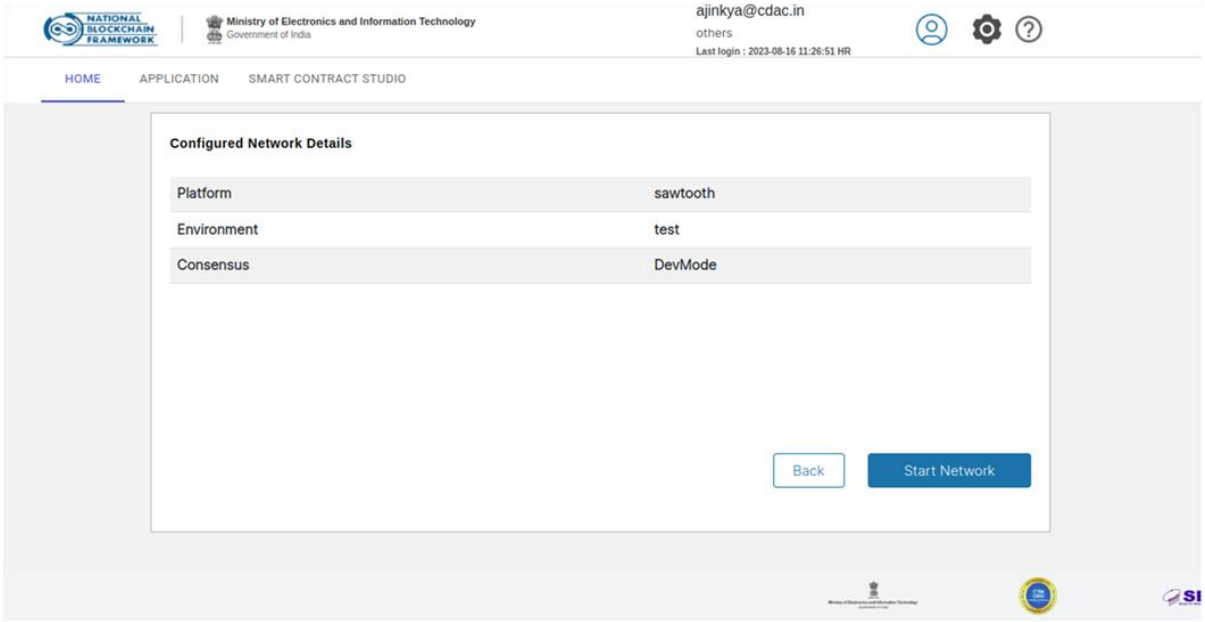


For Sawtooth Network setup, provide the consensus details

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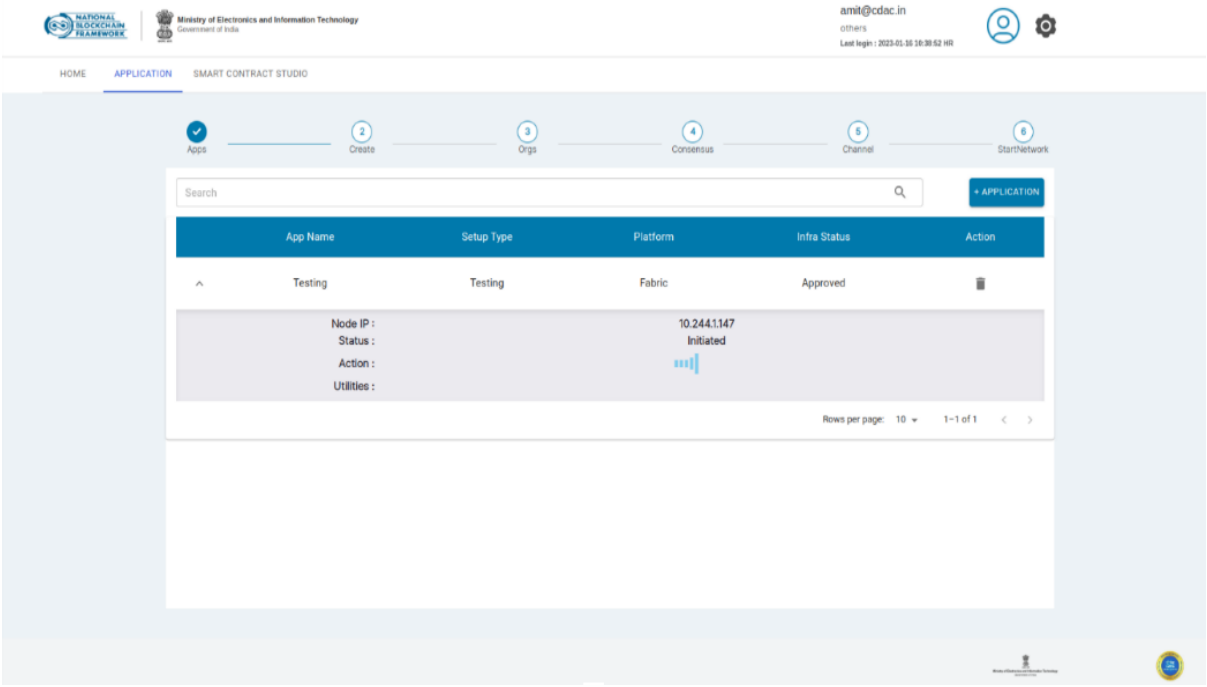


Verify the configuration details and submit for network step up

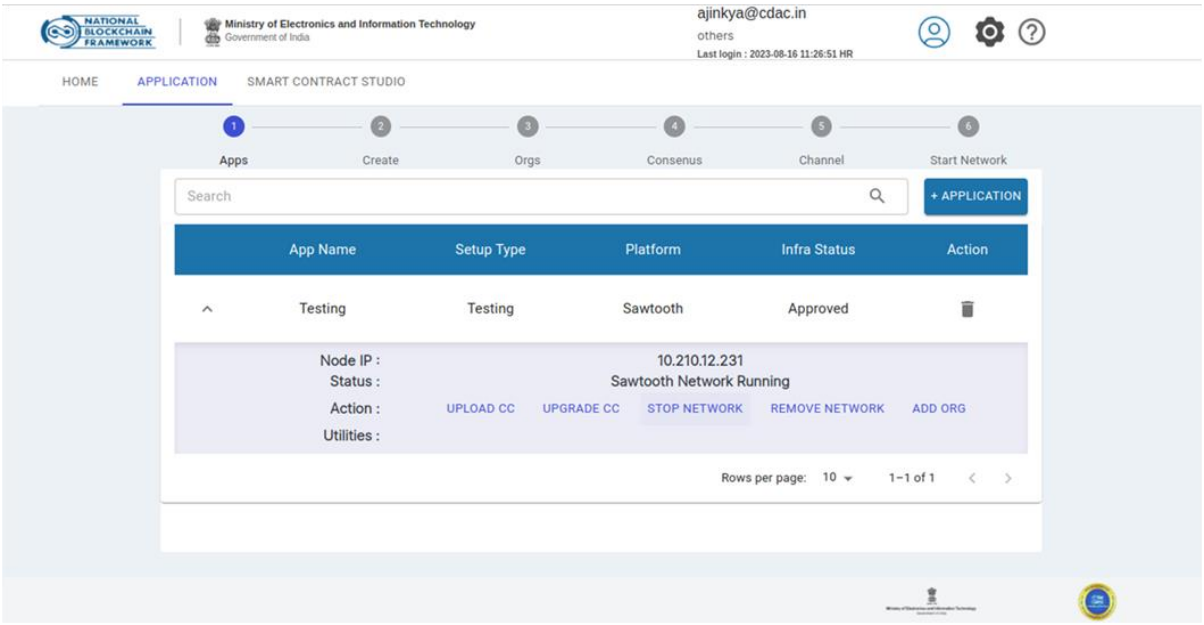


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Network setup is in progress

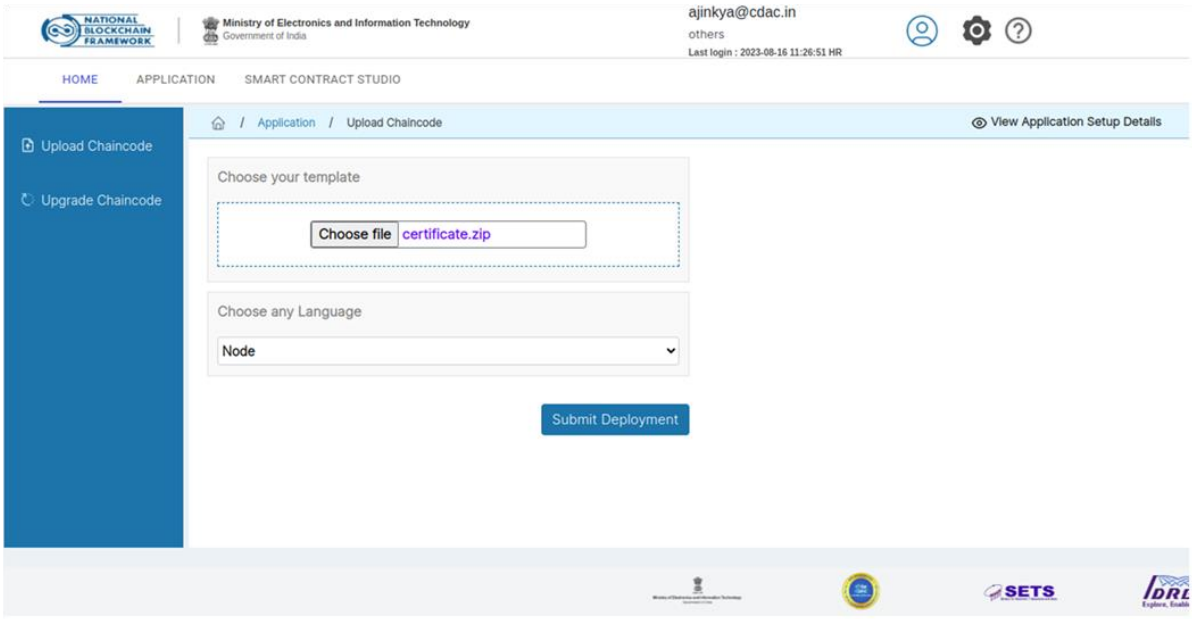


After the network setup,if chaincode already downloaded then upload it using UPLOAD CC else generate chaincode (Refer manual UserManual_SC)

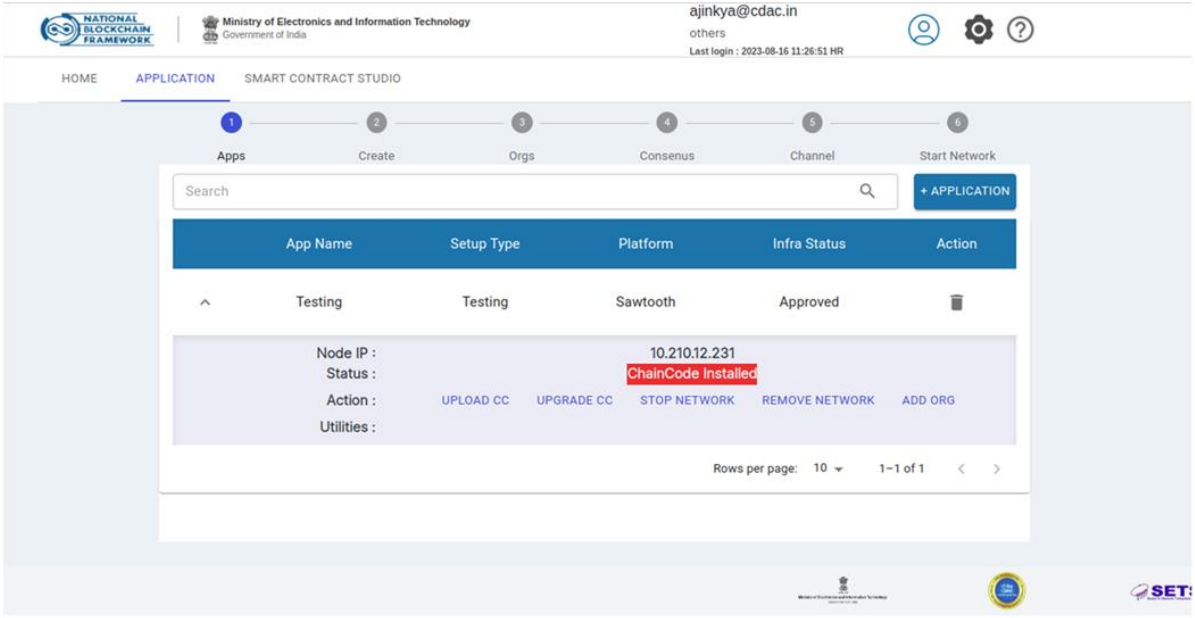


8.2. Deployment of Smart contract

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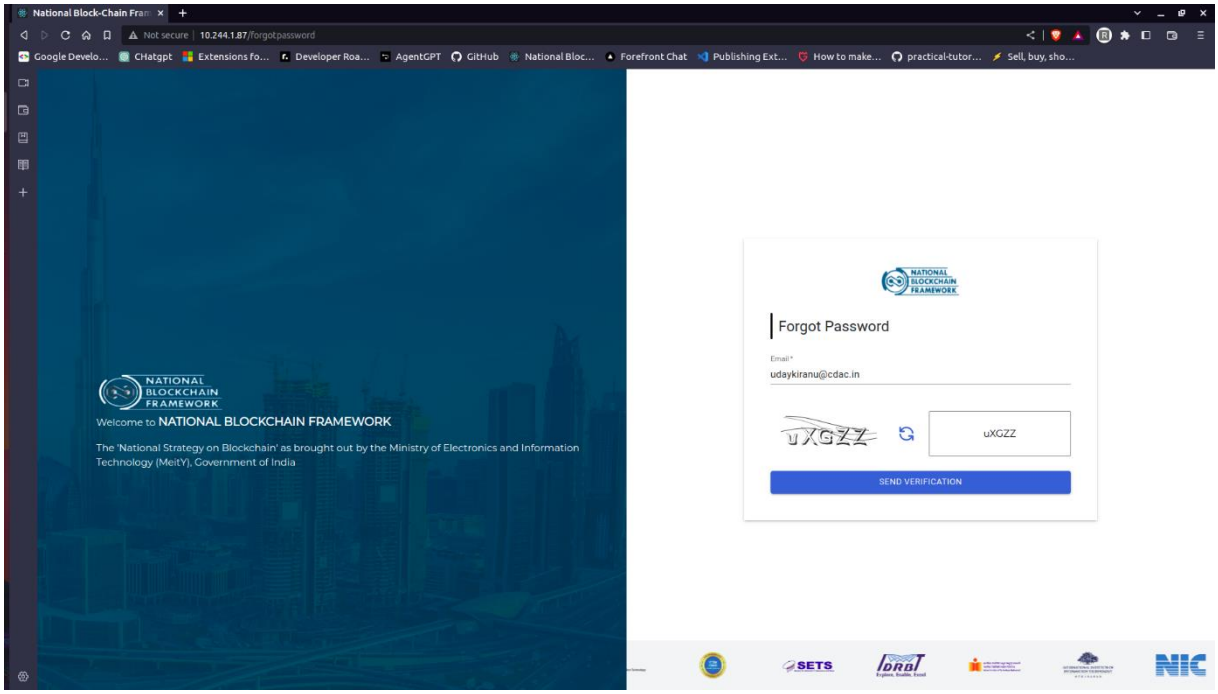


Process completed after the smart contract is deployed



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9. Forgot Password



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E-Mail: Change password for your NBF Portal

Edit Flags

Reply Reply all Forward Delete Print Mark as Save Show source Close

Change password for your NBF Portal

From : cdacchain <cdacchain@cdac.in>

To : udaykiranu@cdac.in

Received : 05-18-2023 12:10 PM

Dear udaykiranu@cdac.in,

Greetings from C-DAC !!!

You have requested password change for your account ID : **udaykiranu@cdac.in** for NBF Portal.

you can now log in to our system using the following credentials

UserName: udaykiranu@cdac.in
Password: %#2E2l$pr

Please note that for security reasons, it is recommended that you change your password after your first successful login.

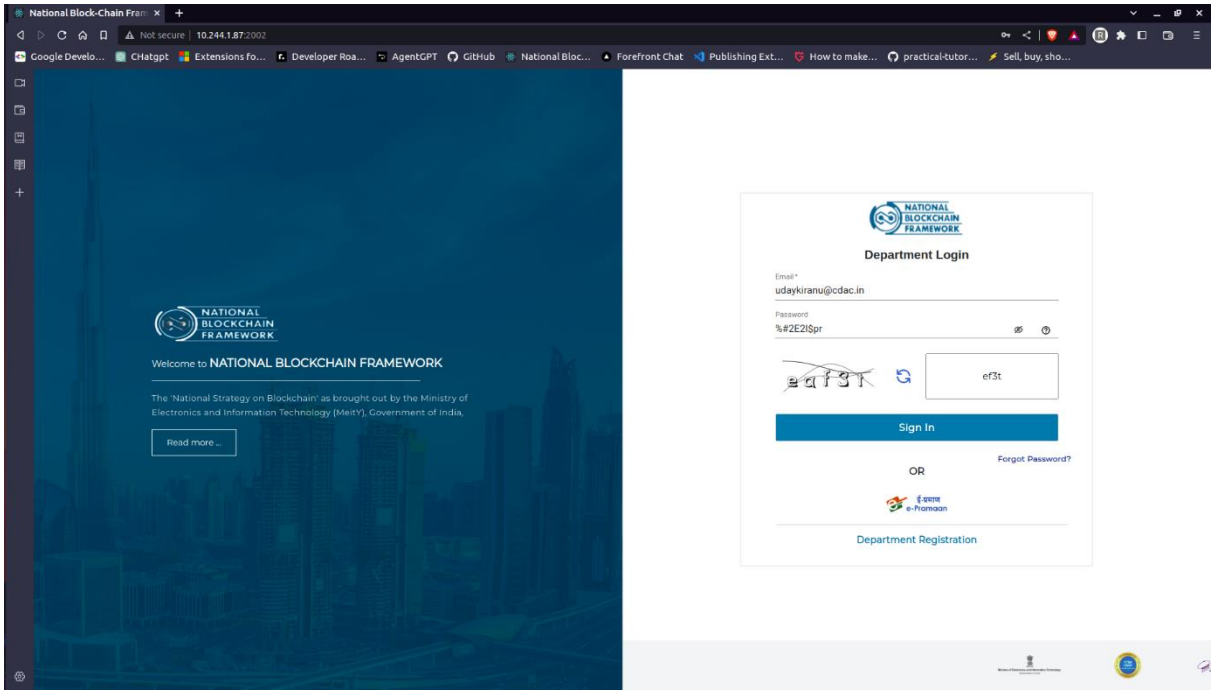
If you have any questions or concerns, please do not hesitate to reach out to our support team.

Your NBF TEAM HYD,

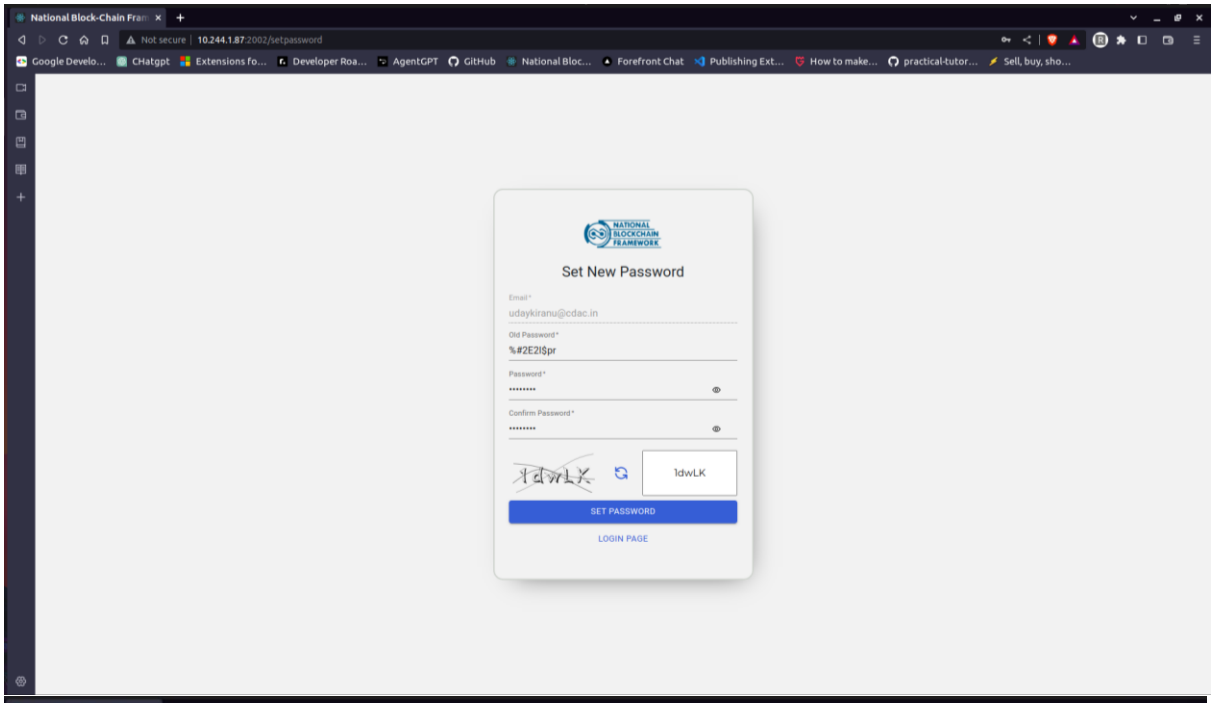
C-DAC, Hyderabad</>

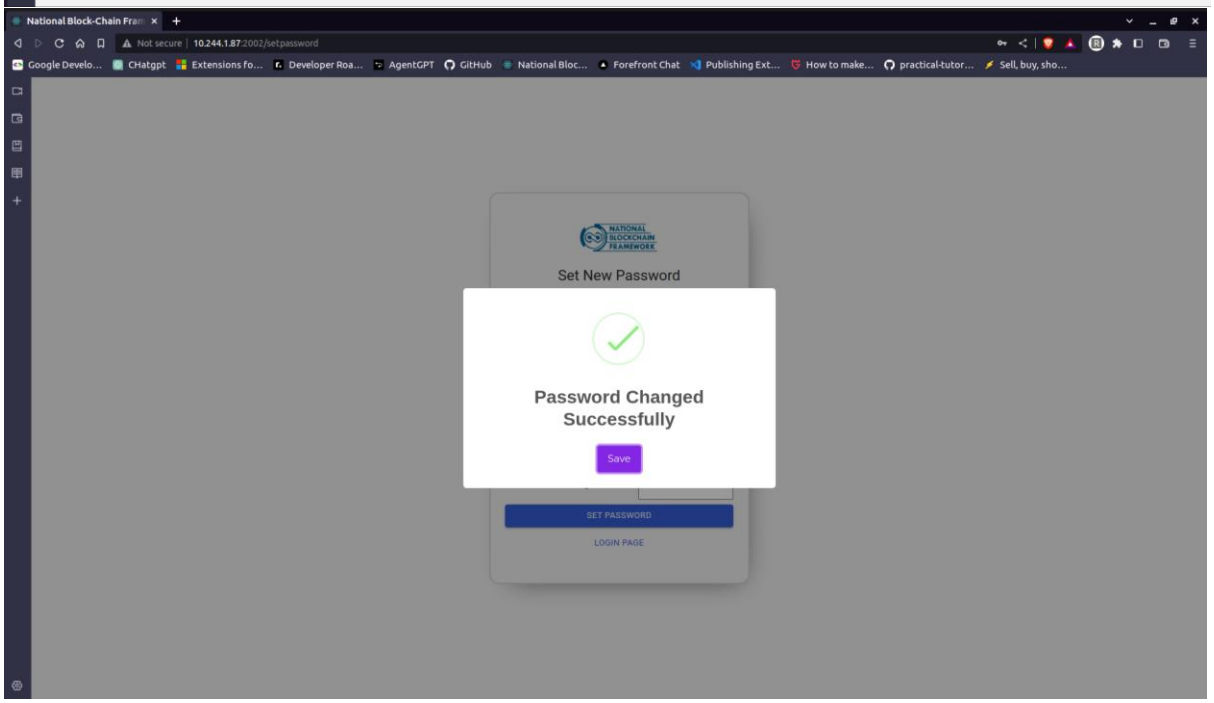
[C-DAC is on Social-Media too. Kindly follow us at:
Facebook: https://www.facebook.com/CDACINDIA & Twitter: @cdacindia]

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